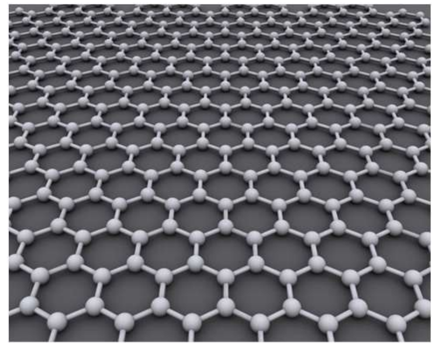
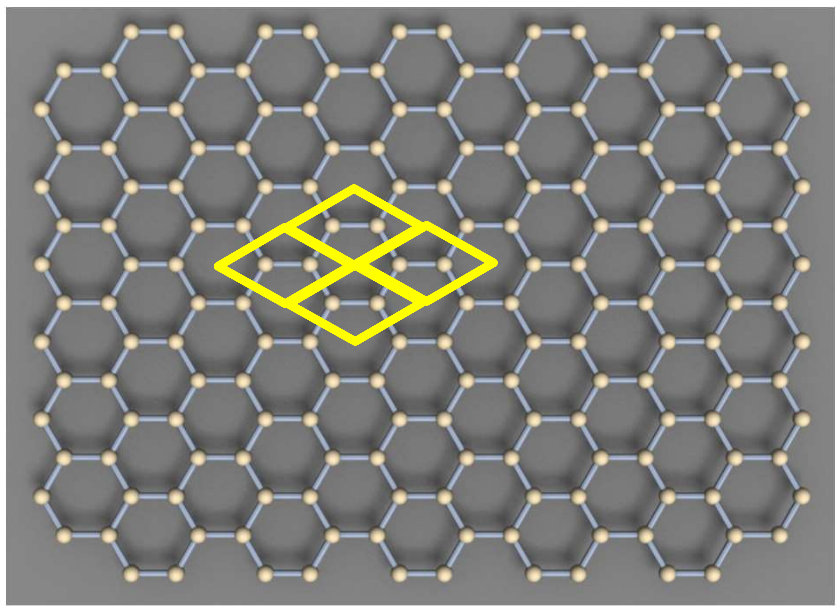
Topic 5: Crystallography in 1D & 2D

Questions we will answer...

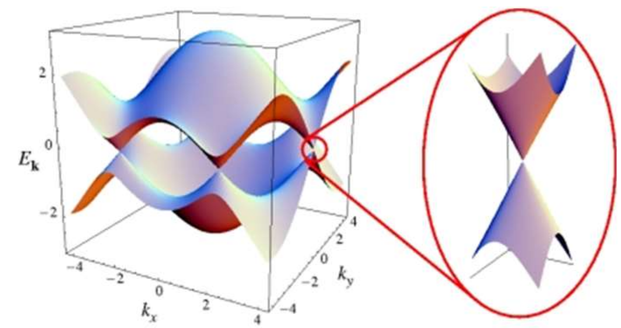
- How do we classify 1D and 2D periodic patterns?
- What are the symmetry elements in 1D and 2D?
- What are the Bravais lattices in 1D and 2D?
- How do symmetry elements combine with a Bravais lattice to create 1D line and 2D plane groups?
- What do plane groups reveal about the symmetry of 2D crystals and how can we interpret them?

2D Nanomaterial: Graphene

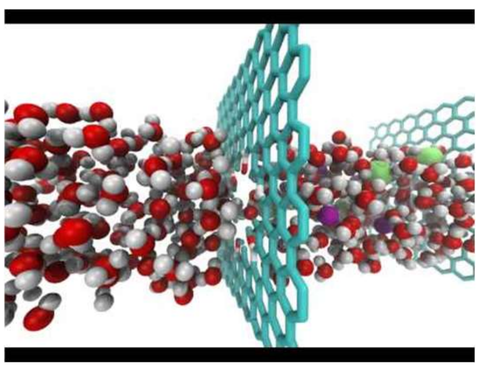




How do we classify 2D crystals and surfaces?
We will learn about 2D unit cells, lattice points, the basis.
We will learn about symmetry elements, symmetry operations, point groups and plane groups…



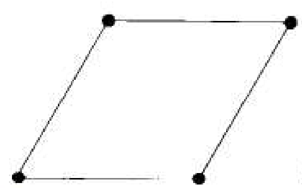
http://condensedconcepts.blogspot.com/2013/04/the-dirac-cone-in-graphene-is-emergent.html



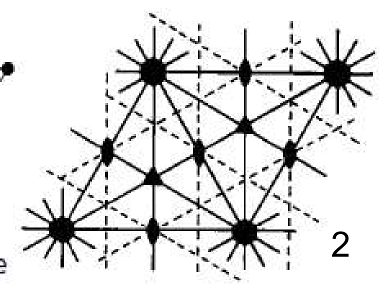
https://www.youtube.com/watch?v=EOVduH-iGMw

Plane group: p6mm

What does this code tell us about graphene?
How are 2D material structures classified?



The hexagonal p-lattice



$p6mm$

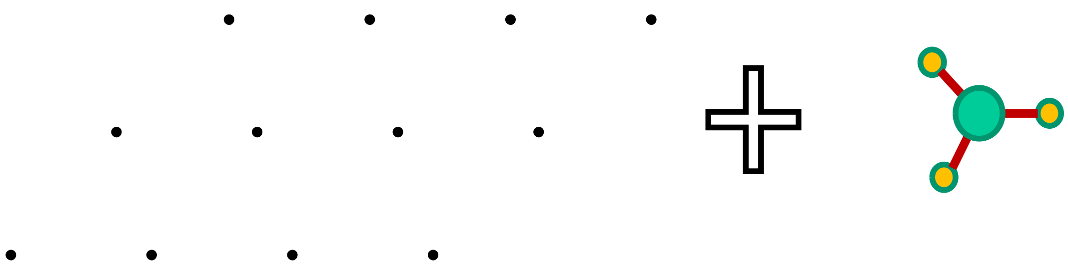
What Is Crystallography?

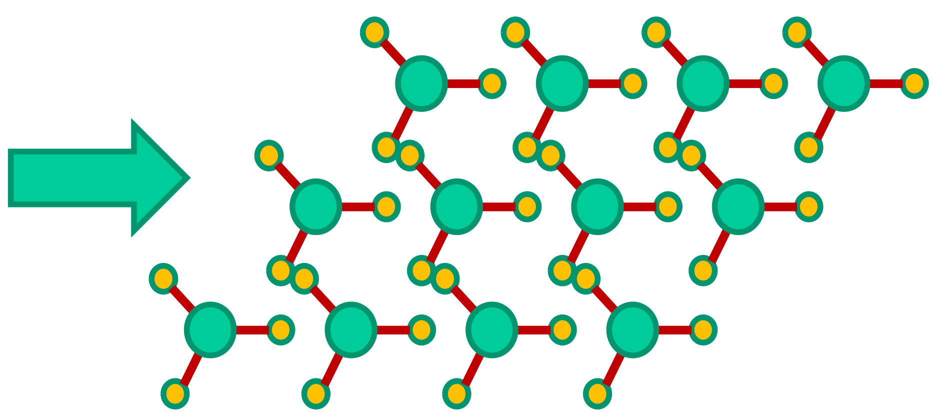
Crystals

Atoms that are bound together ➔ minimize the energy

➔generally achieved by a periodic arrangement

➔ Two elements: lattice + basis





In 1D…

1 Bravais lattice

In 2D…

5 Bravais lattices

In 3D…

14 Bravais lattices

In 1D…

6 Line groups

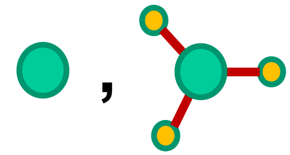
In 2D…

17 Plane groups

In 3D…

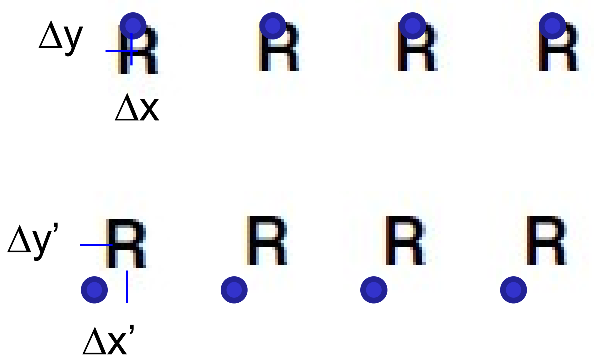
230 Space groups

Unit cells and lattices in 1D

R examples:  **, …**

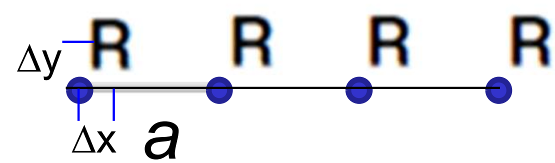
Unit Cell

Assume an infinite 1D linear array of **R**s; the **R** is a repeating pattern, known as a **basis**, on a lattice with a unit cell parameter *a*



Lattice points: anywhere will do!

Only requirement: position of lattice relative to basis must be known and maintained.



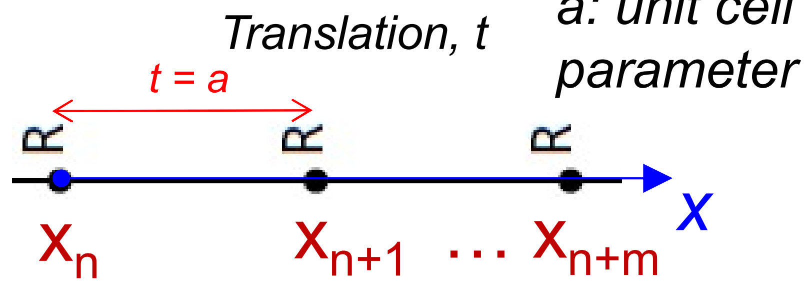
If we know the length of the unit cell, the basis and its position/orientation within it, then we can reproduce the structure completely and indefinitely through translation (and other symmetry operations)

∴the <u>unit cell</u> and <u>basis</u> define the entire structure of a crystal in 1D and encode all of its details, symmetries, etc.

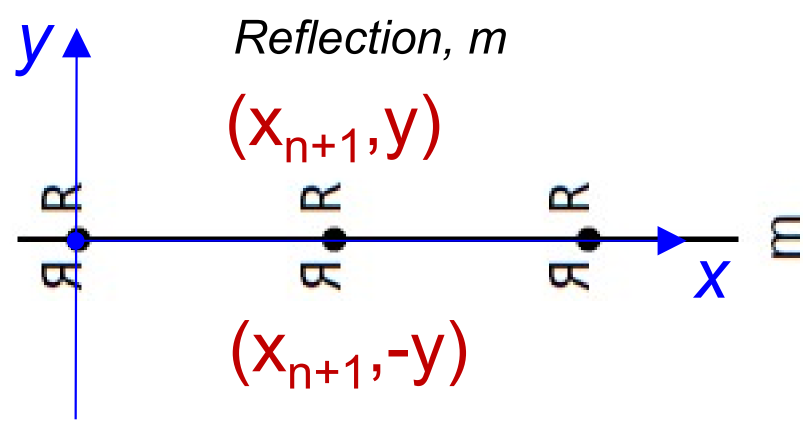
Symmetry Elements and Operators

A **symmetry operator** describes the imaginary action of a symmetry element placed at a specific location to develop a pattern (*e.g. lattice point + translation; lattice point + diad (rotation); lattice point + mirror (reflection)*)

- Operators change the position/orientation and handedness of an object in space
- Each operator can be described mathematically



$$x_{n+m} = x_n + mt$$



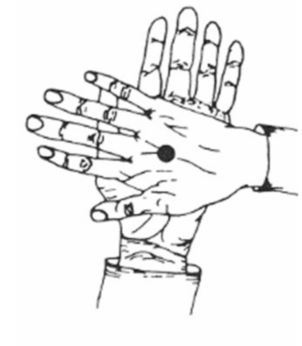
Seven symmetry elements in crystallography:

- **Translation** (1D, 2D, 3D)
- **Rotation** (1D, 2D, 3D)
- **Reflection** (mirror) (1D, 2D, 3D)

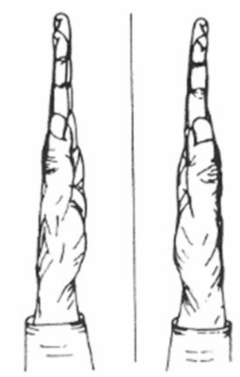
Topic 5

- **Inversion** (center of symmetry) (3D)
- **Roto-inversion** (inversion axis) (3D)
- **Glide** (translation + reflection) (2D, 3D)
- **Screw** (rotation + translation) (3D)

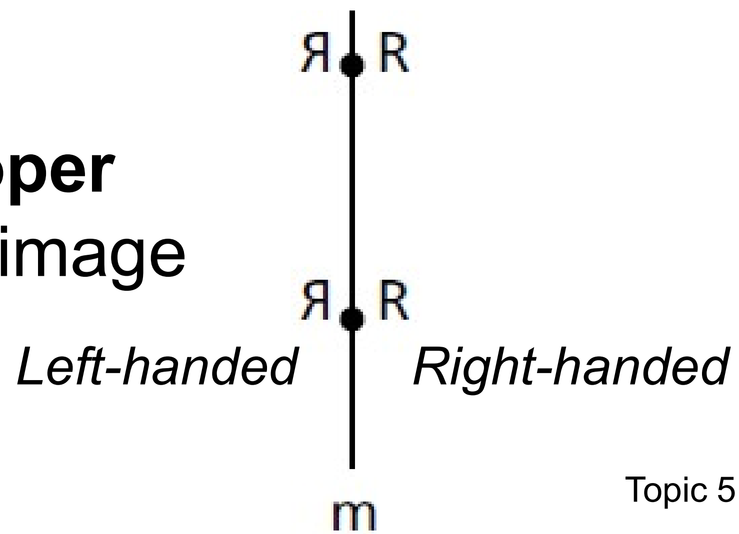
Proper vs. Improper Operations



- *Proper*: operation that produces a replica with same handedness (R → R)
- *Improper*: operation that produces a left-handed replica (R → Я)
- Left-handed and right-handed objects cannot be superimposed by any combination of a rotation or translation (proper operations)

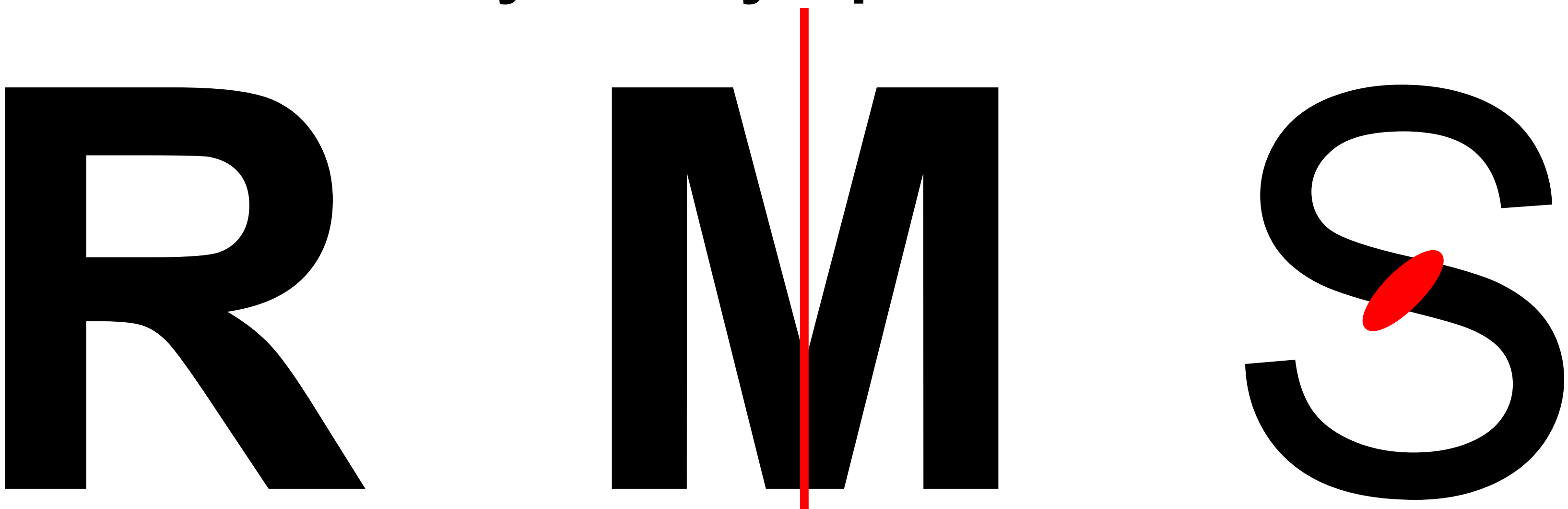


Reflection (mirror) is an **improper** symmetry operation (creates an image with *opposite congruence*)



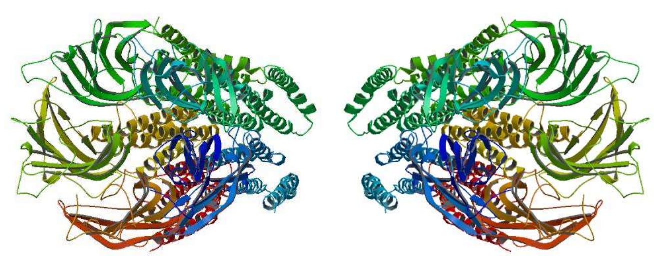
Symmetry Elements & Bases

Symmetry Operators



Mirror line
of symmetry

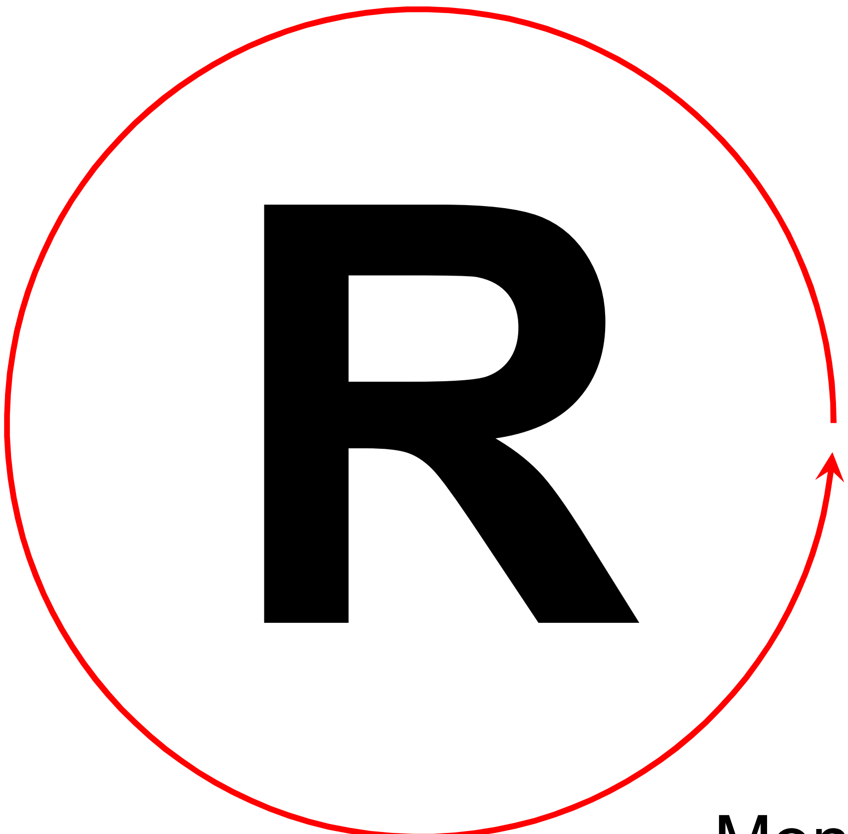
2-fold rotation (180°)
a.k.a. *diad axis*



A basis can have symmetries

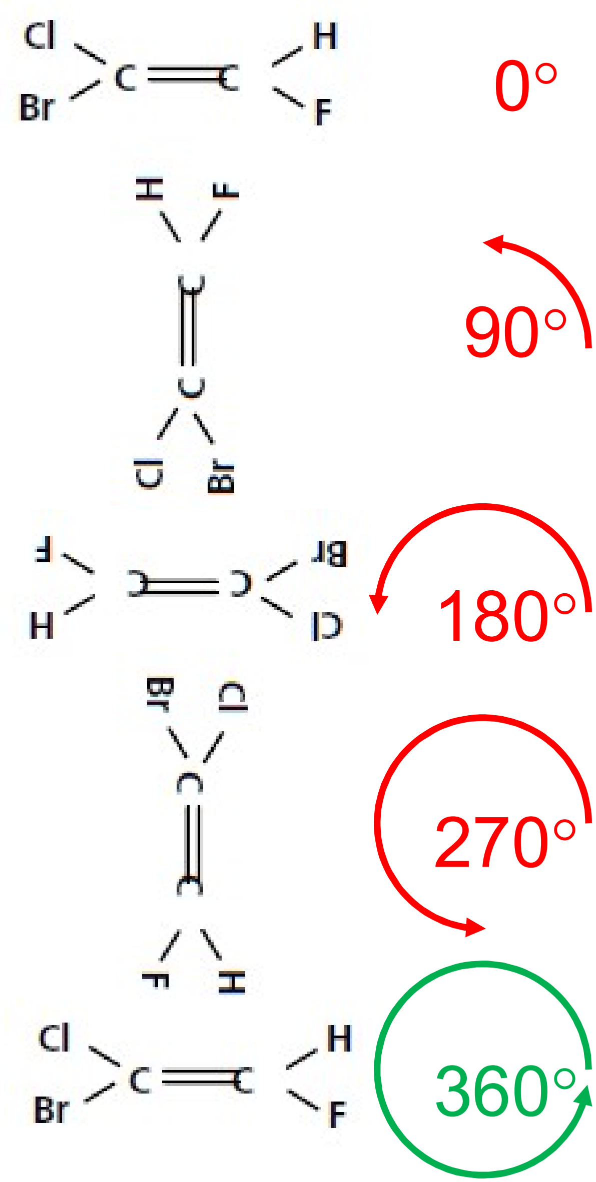
R is any *asymmetric basis*… a molecule, a cluster of atoms; compare with **M** and **S**!

Symmetry Elements & Bases



360° or 2π

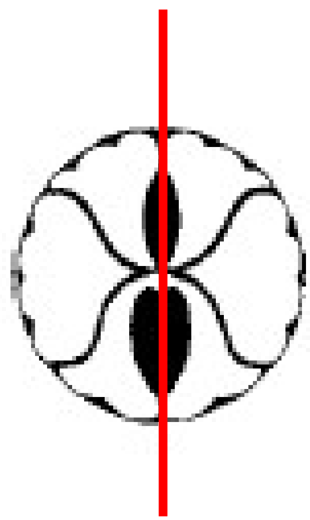
Monad



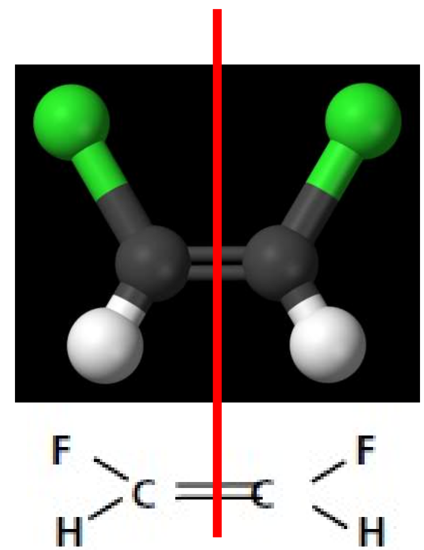
bromofluorochloroethene

Point group: **1**

Symmetry Elements & Bases



Monad

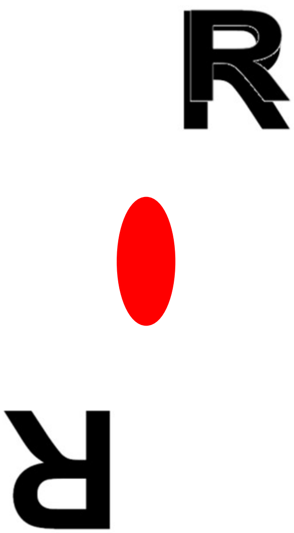


Cis-difluoroethene

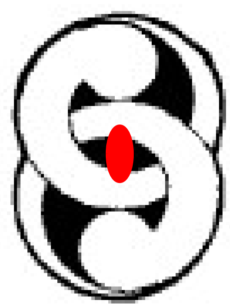
Left-handed **R** Right-handed **R**

Point group: **m**

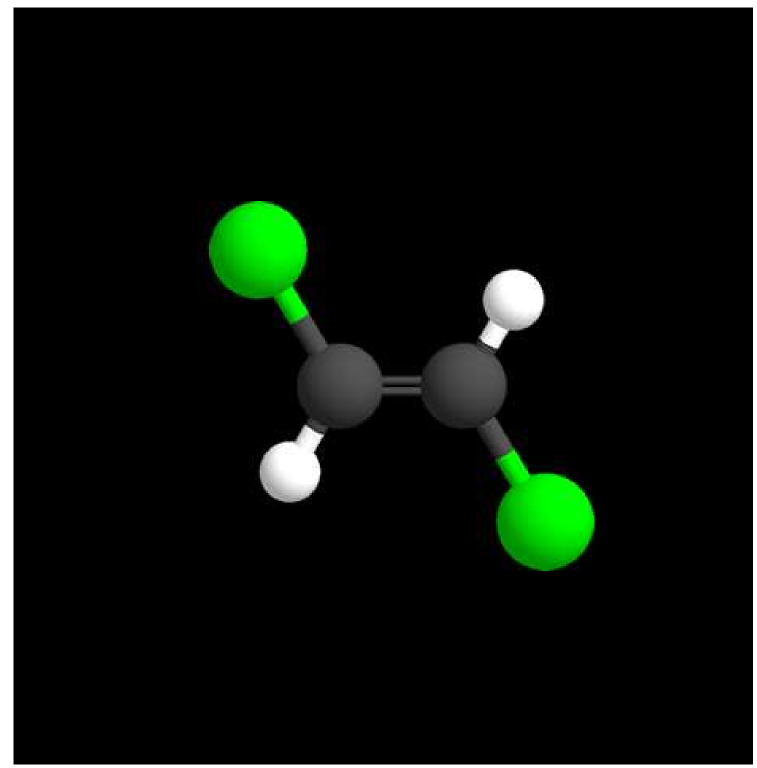
Symmetry Elements & Bases

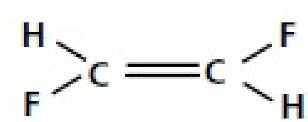


Diad



Same image repeated every 180° or π rotation

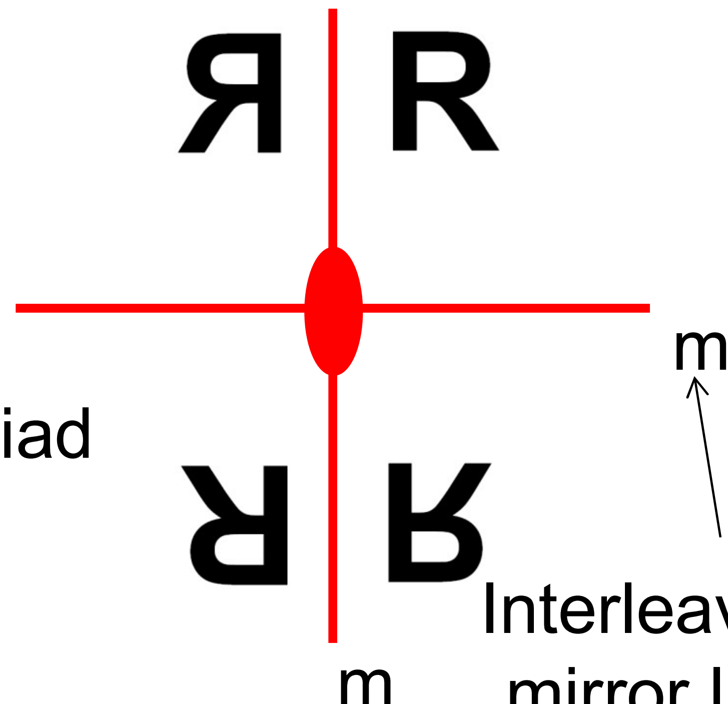




Trans-difluoroethene

Point group: **2**

Symmetry Elements & Bases

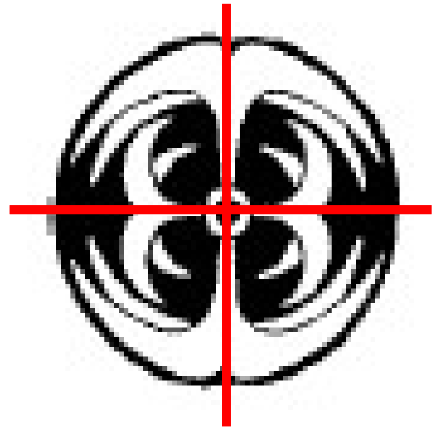


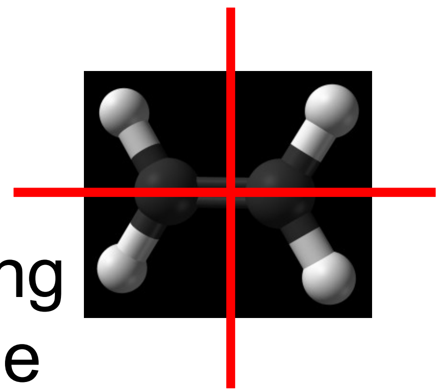
Diad

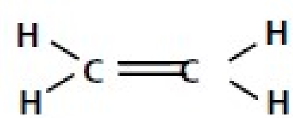
Interleaving mirror line

Combination of diad and mirrors

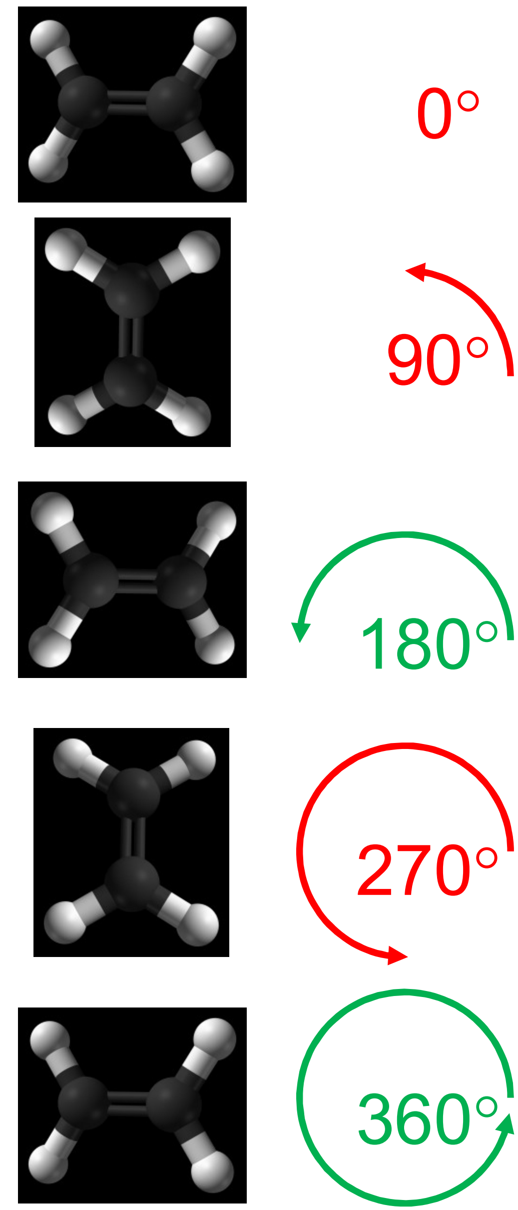
Point group: **2mm**





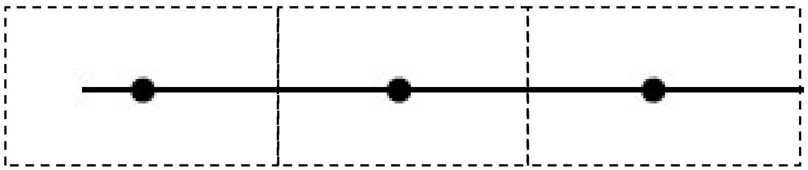


ethene



Constructing a 1D Lattice

1D lattice (translation)



Unit cell

+

4 Point group symmetries

1, *m*, 2
2mm

Lattice is primitive (p):
One point per unit cell

One type of 1D lattice: Supports all 4 point group symmetries

=

6 Line groups

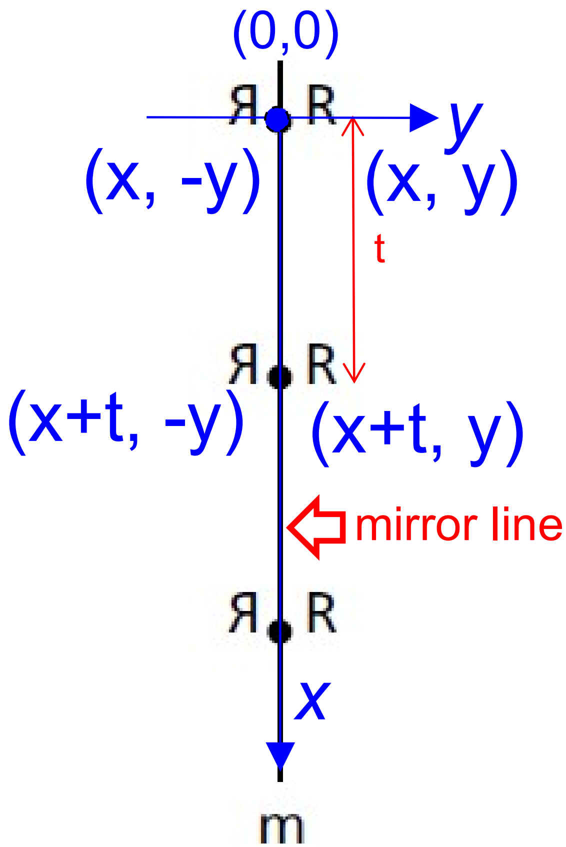
p1, pm, p2, pg,
p2mm, p2mg

Line group encodes all *essential* symmetries, i.e., symmetries required to replicate the lattice and basis in 1D

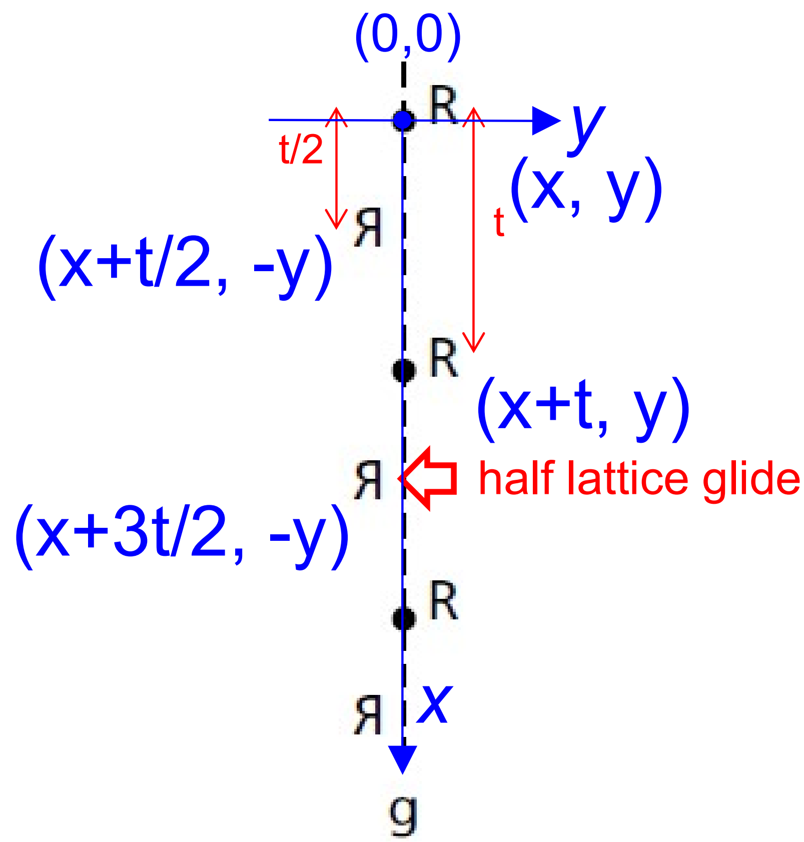
Symmetries of line (plan/space) group are determined jointly by symmetries of lattice AND basis

Reflection vs. Glide-Reflection Symmetry

Reflection symmetry



Glide-reflection symmetry
a.k.a *reflection glide* or *glide line of symmetry*



Basis has a mirror symmetry

Basis has <u>no</u> mirror symmetry

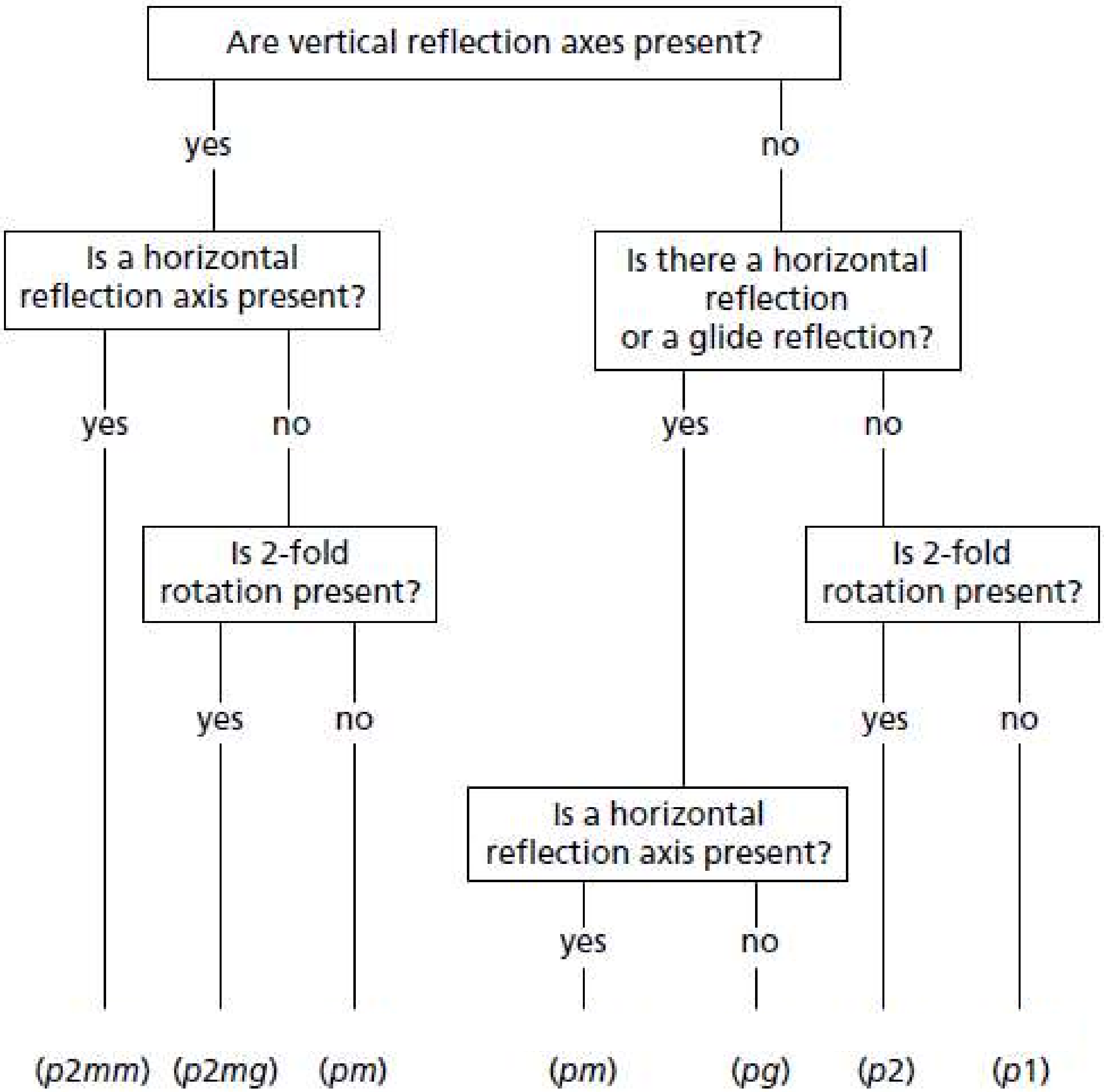
Я R

Glide = translation
+ reflection
Line groups: pg & p2mg

R

Я

1D Groups - Flowchart

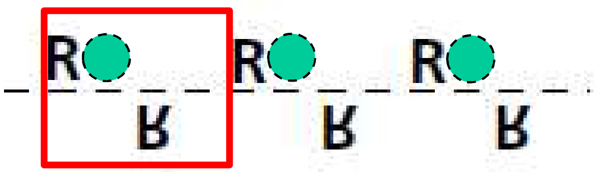


Six unique Line Groups:
p1, p2, pm, pg, p2mg, p2mm

1D Groups

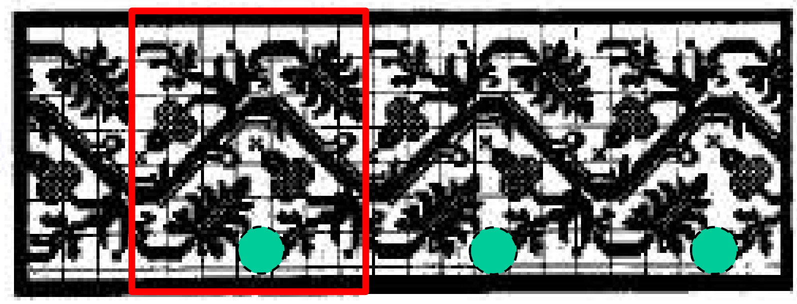
Lattice + monad

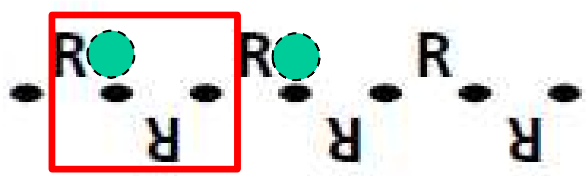
p1



Lattice + glide

pg



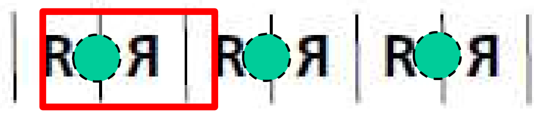


Lattice + diad

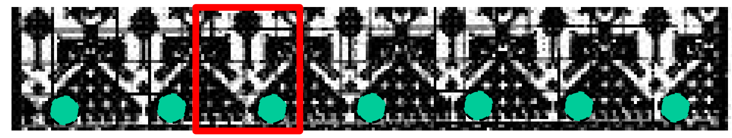
p2

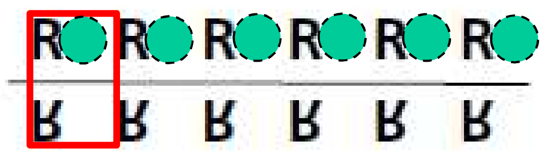
Unit cell, lattice? *What about the basis?*

1D Groups

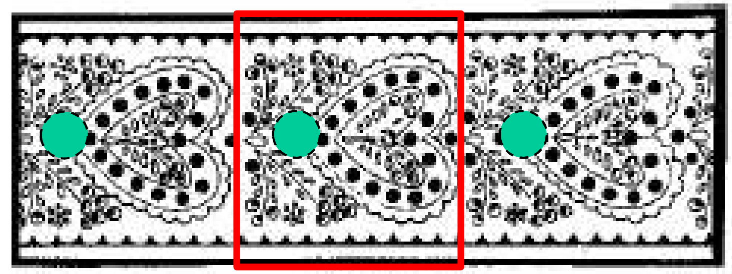


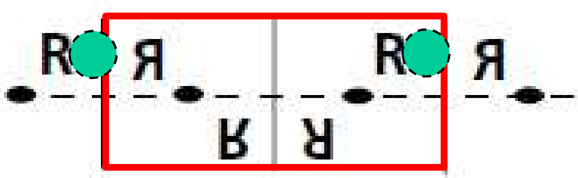
pm



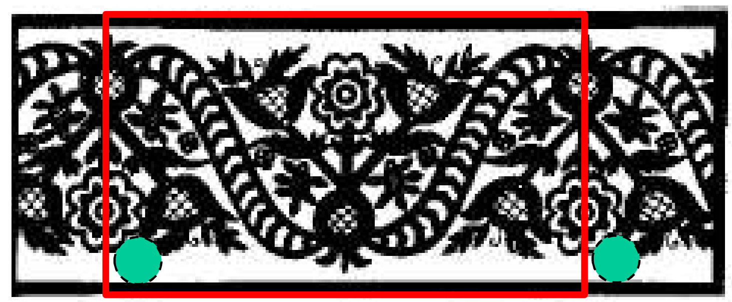


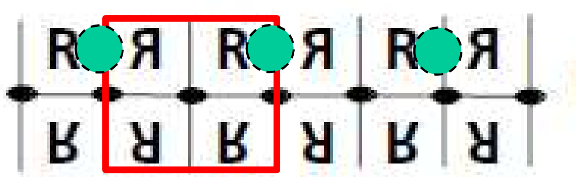
pm



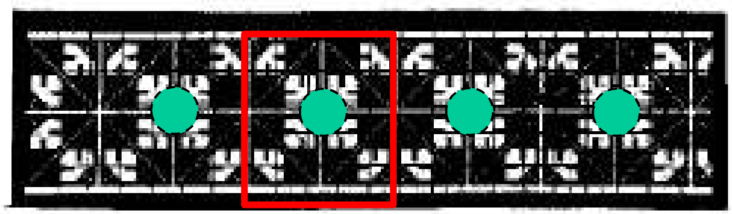


p2mg





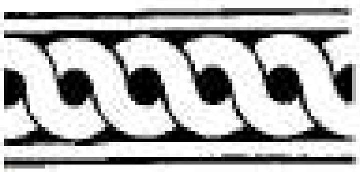
p2mm



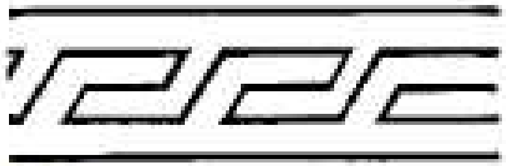
Unit cell, lattice? *What about the basis?*

In-Class Exercise 1

Find the unit cell, lattice, symmetry elements, and line group.



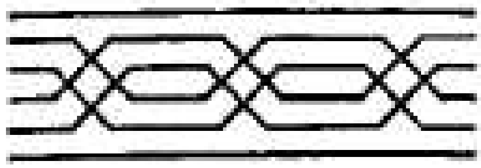
(a)



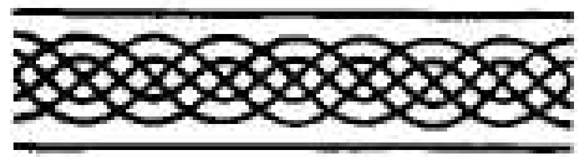
(b)

(c)

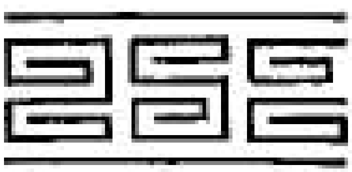
(d)



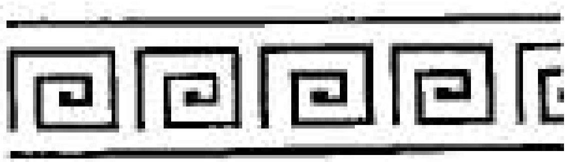
(e)



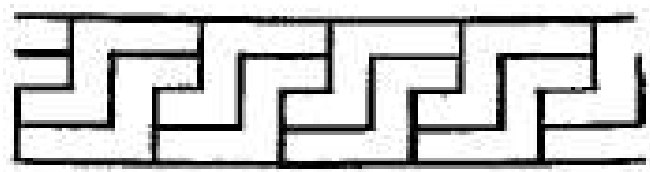
(f)



(g)



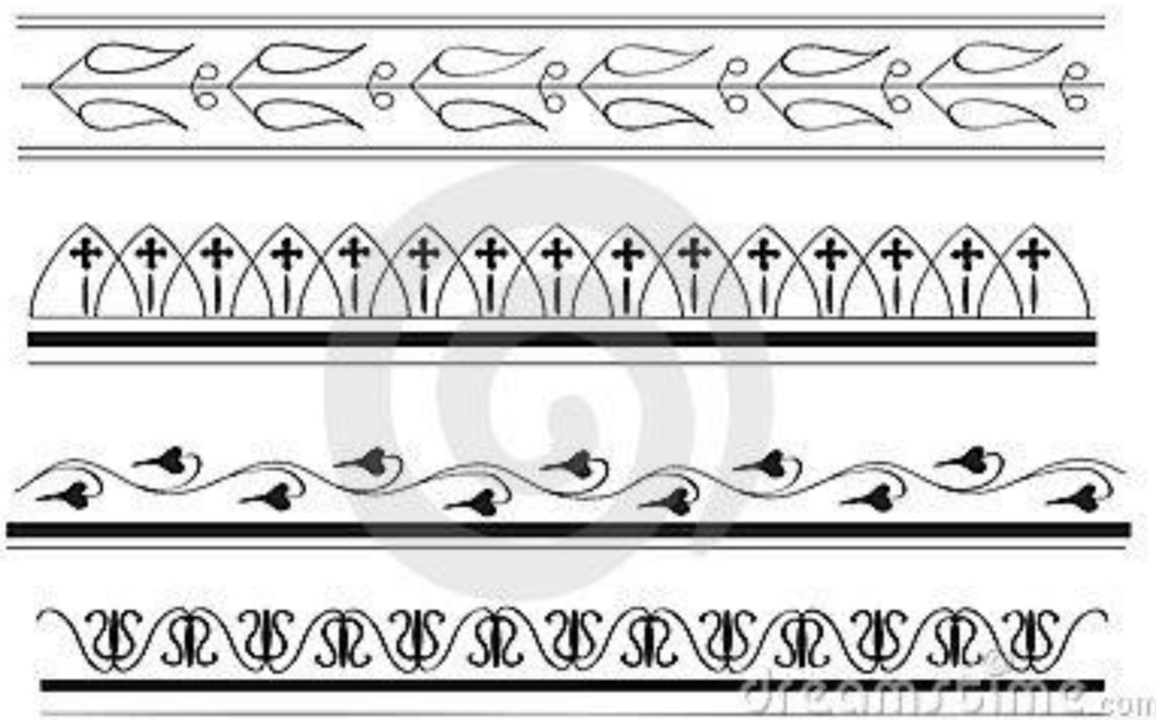
(h)



(i)

In-Class Exercise 2

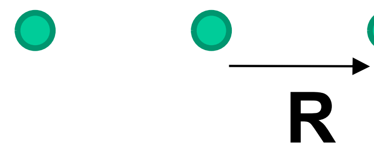
Identify the unit cell, as many symmetry elements as you can and the point group of the following structures.



Rotational Symmetry Elements in 2D & 3D Bravais Lattices

Symmetry Operators

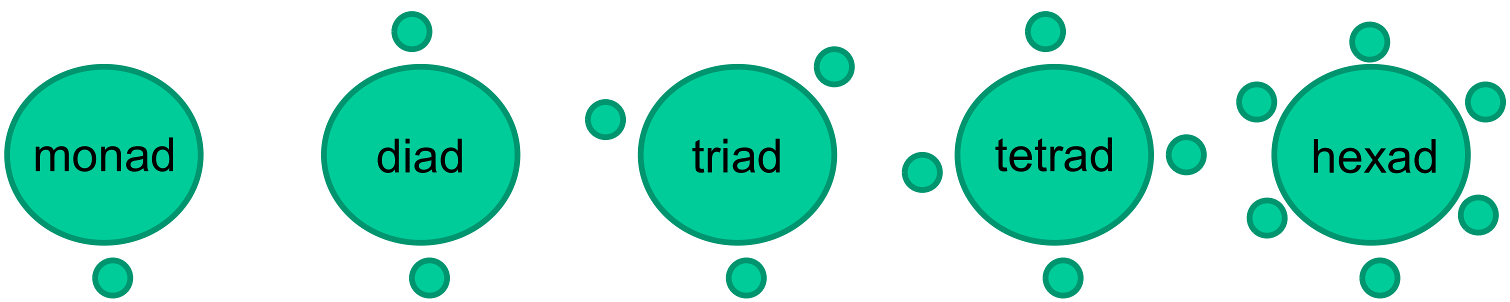
- **Translation**



$$\mathbf{R} = x\mathbf{a} + y\mathbf{b}$$

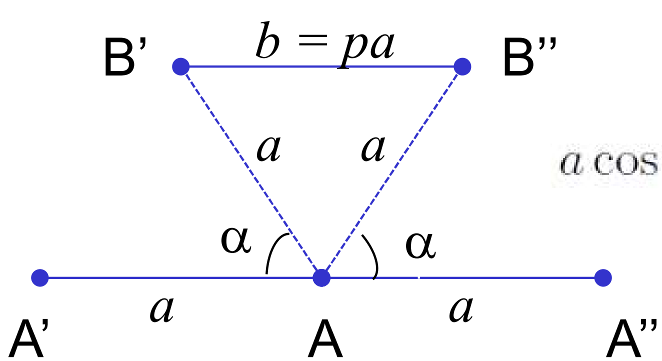
- **Rotation**

$$n\alpha = 2\pi, \;\; n = 1, 2, 3, 4, 6$$



Compatibility between rotations and translation

➔ We are only interested in Bravais lattices!



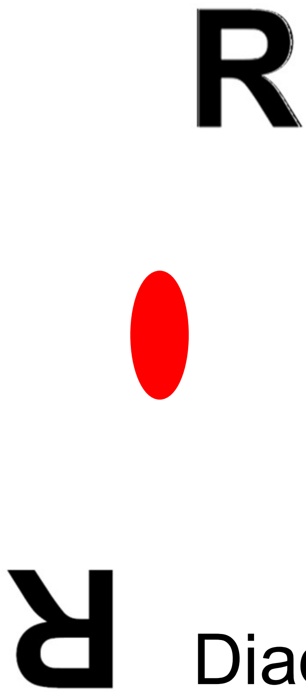
$$a \cos\alpha = \frac{b}{2} \Rightarrow \cos\alpha = \frac{p}{2}$$

$1+2\cos(2\pi/n)$ = integer

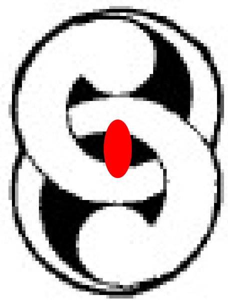
n = 1, 2, 3, 4, 6

are the only *valid* rotational symmetries (5).

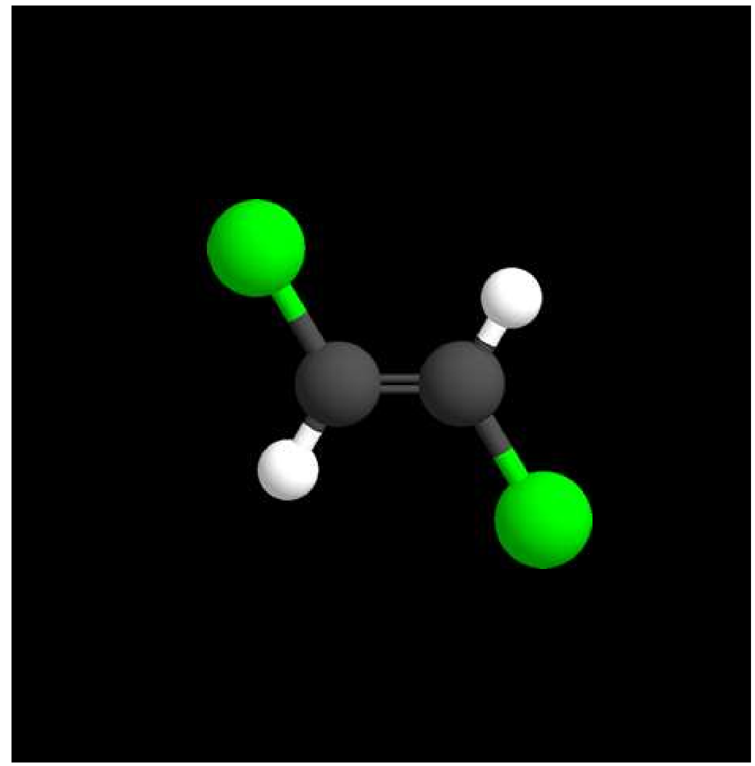
Symmetry Elements in 2D

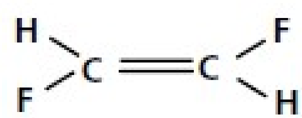


Diad



Same image repeated every 180° or π rotation

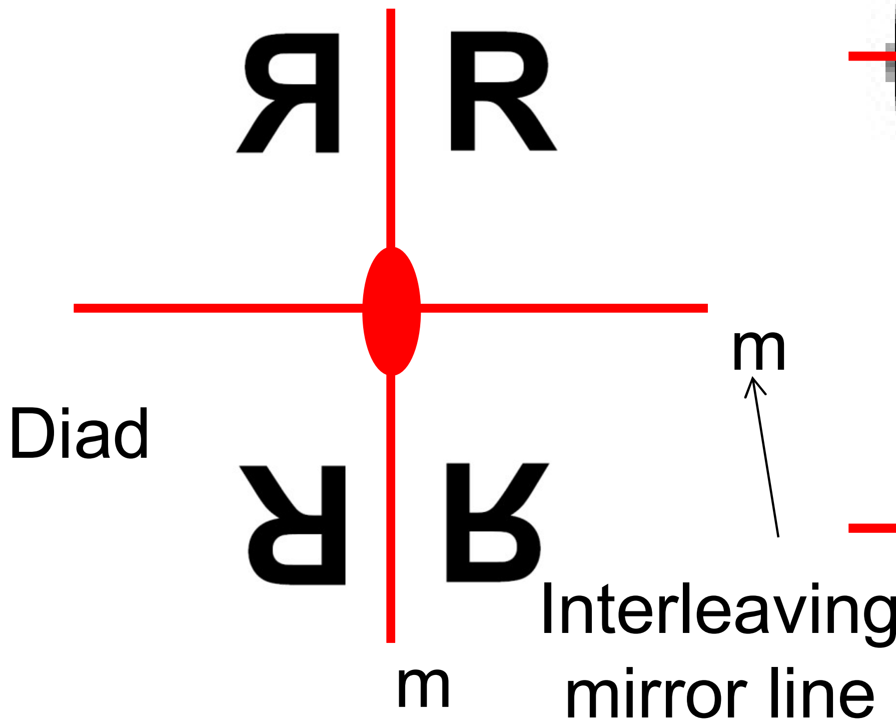


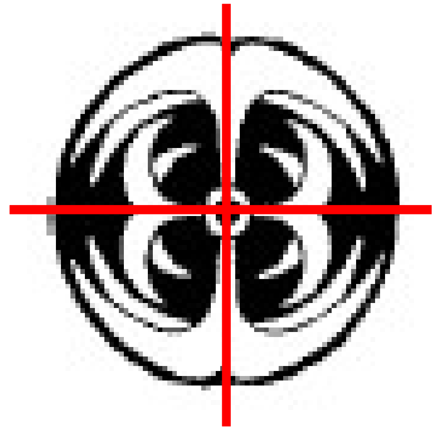


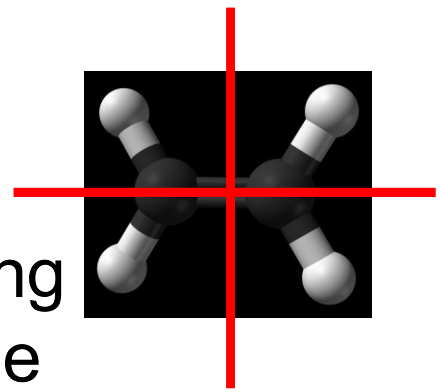
Trans-difluoroethene

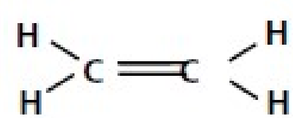
Point group: **2**

Symmetry Elements in 2D

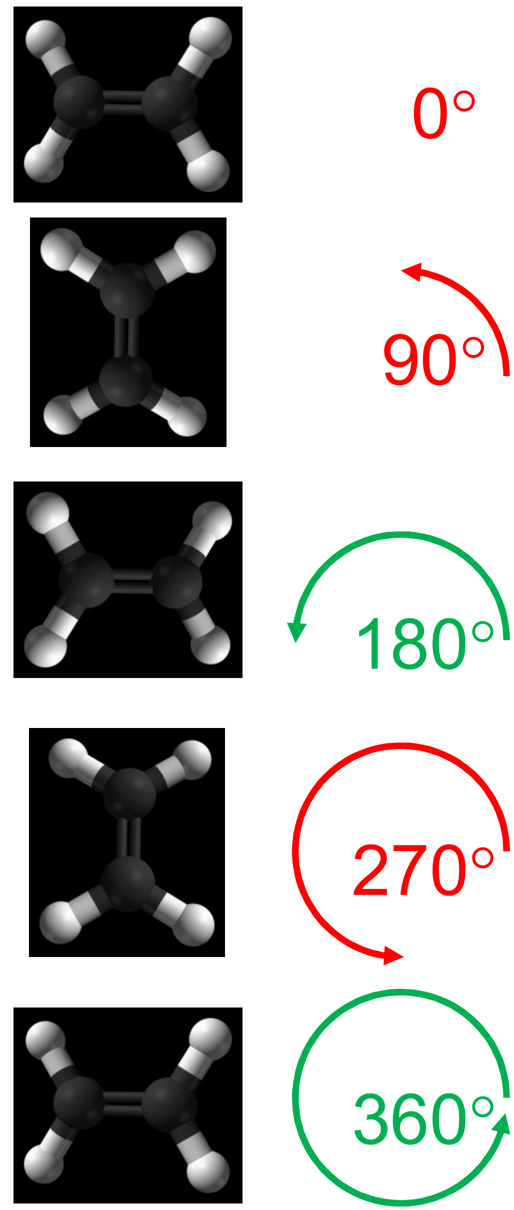








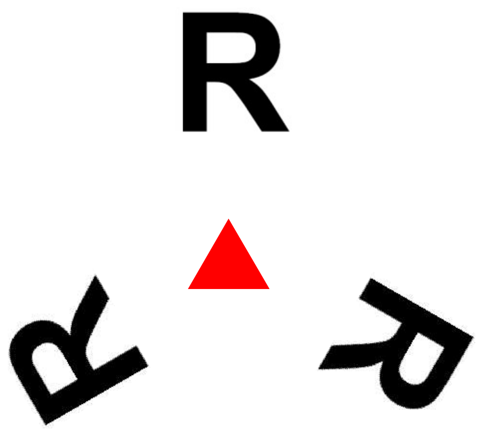
ethene



Combination of diad and mirrors

Point group: **2mm**

Symmetry Elements in 2D



Triad

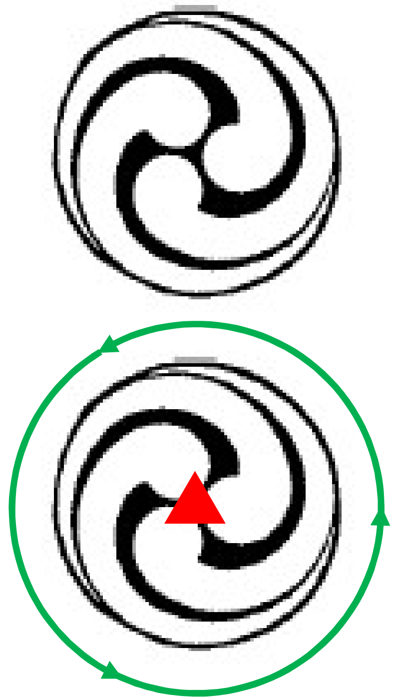
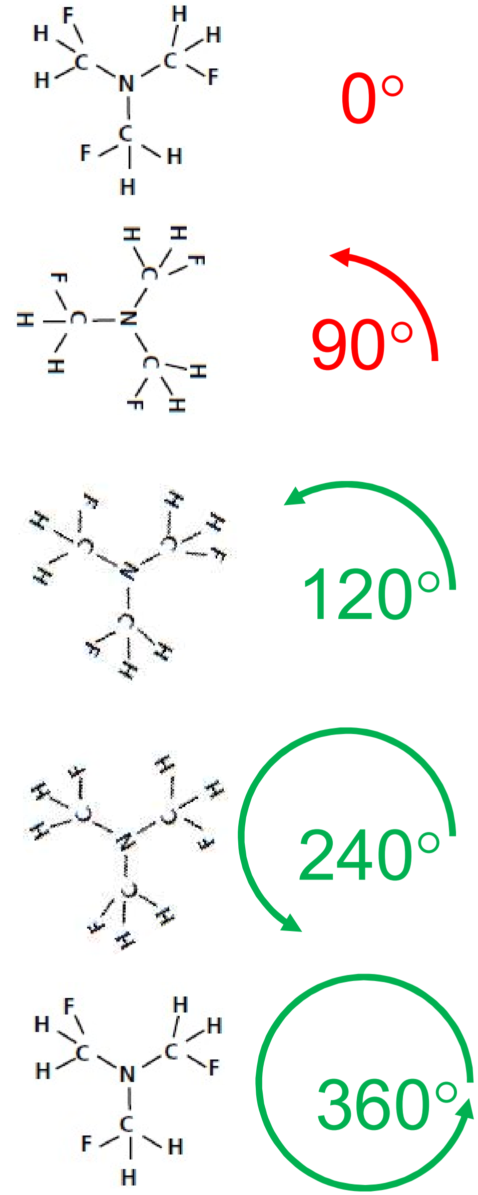


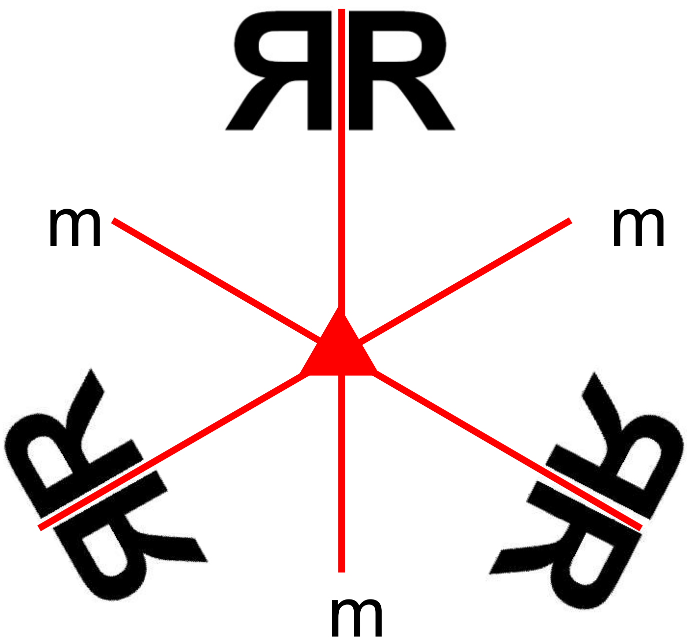
Image repeats itself
3-times per rotation,
or every
120° or $2\pi/3$

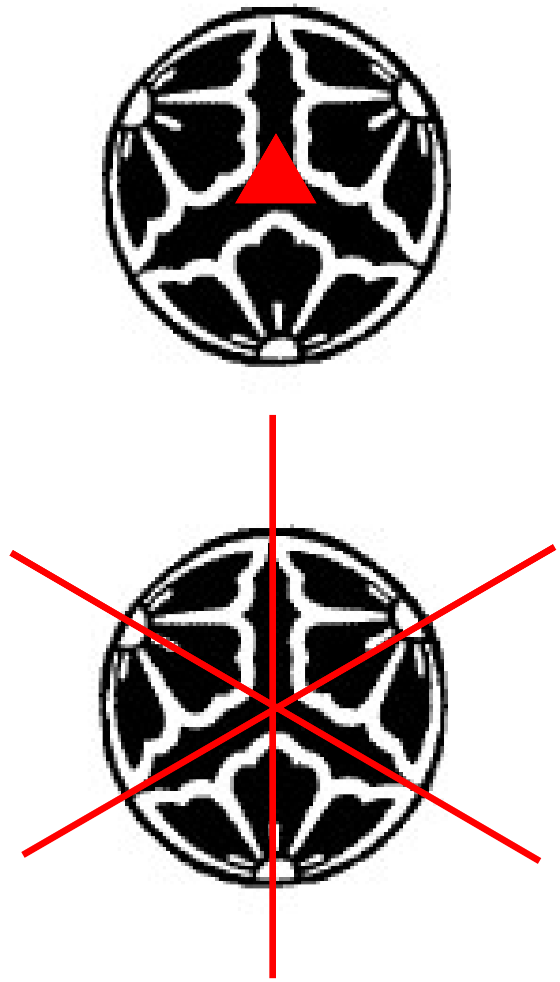


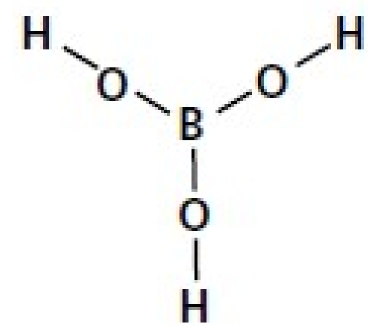
trifluoroalkylammonia

Point group: **3**

Symmetry Elements in 2D





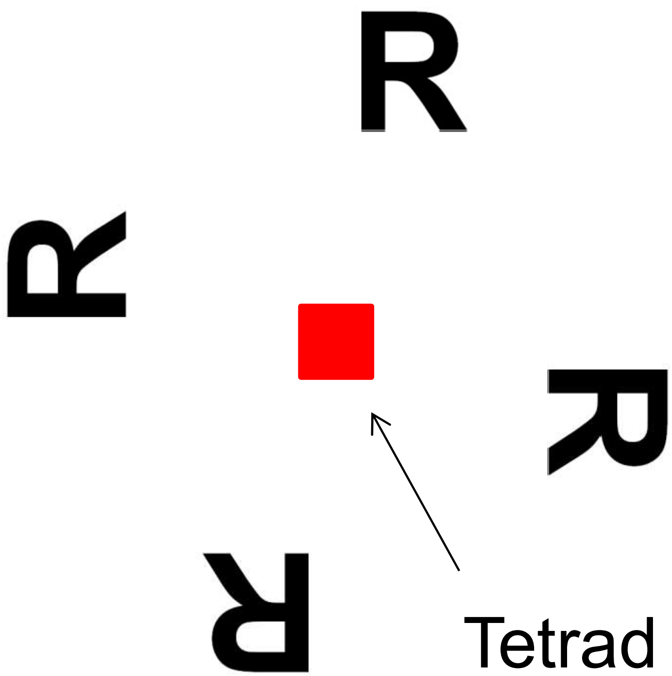


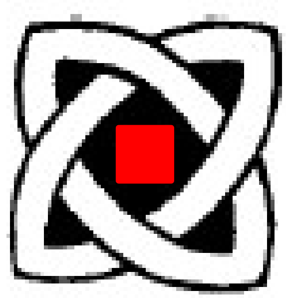
Boric acid

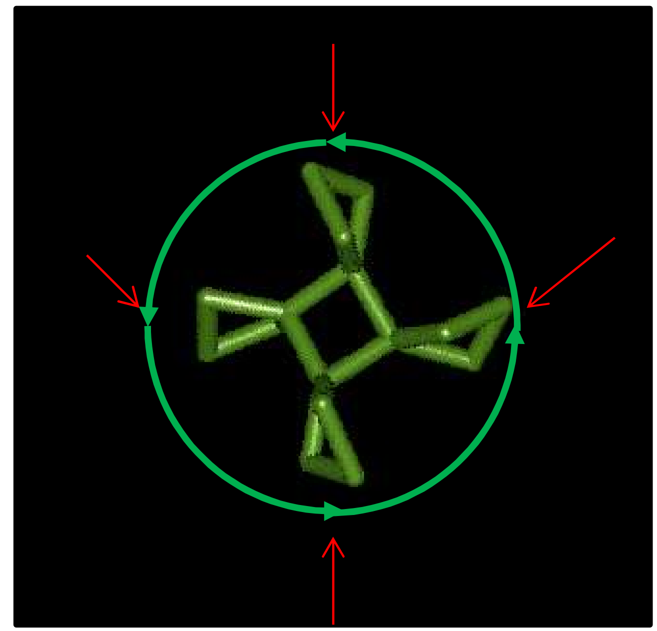
Combination of triad and mirrors

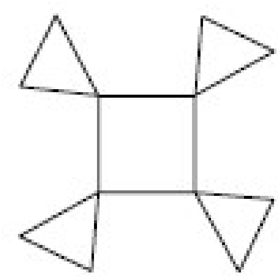
Point group: **3m**

Symmetry Elements in 2D





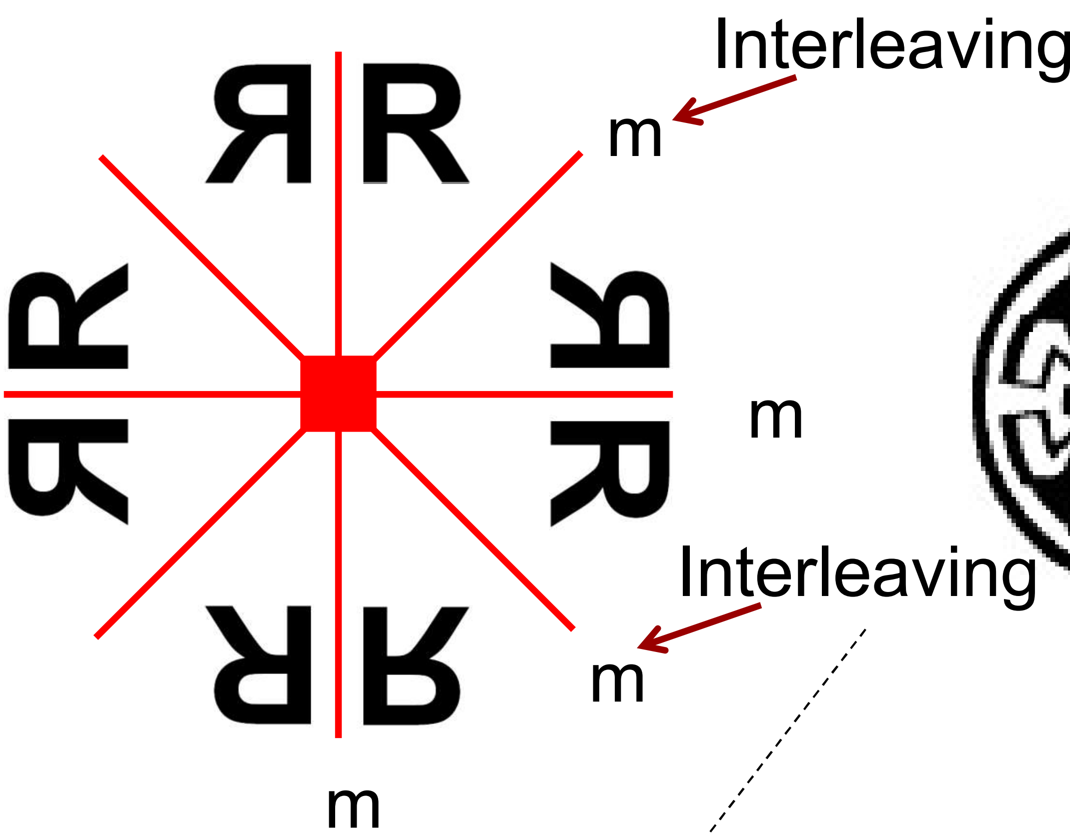




4-rotane ($C_{12}H_6$)

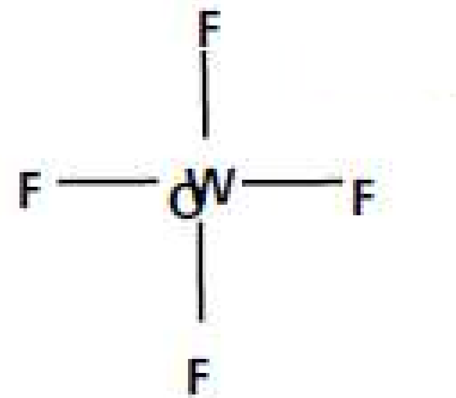
Point group: **4**

Symmetry Elements in 2D



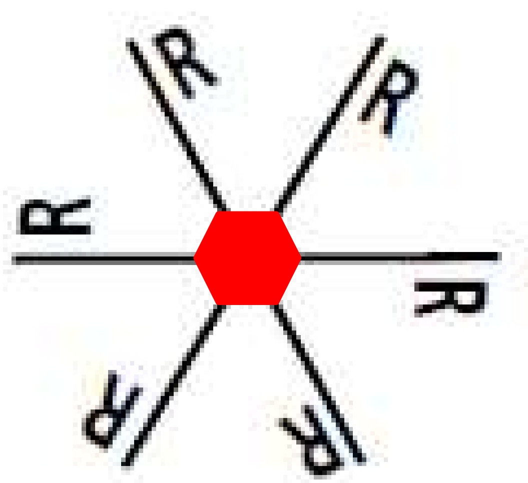
Combination of terad and mirrors

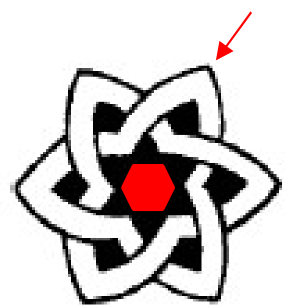
Point group: **4mm**

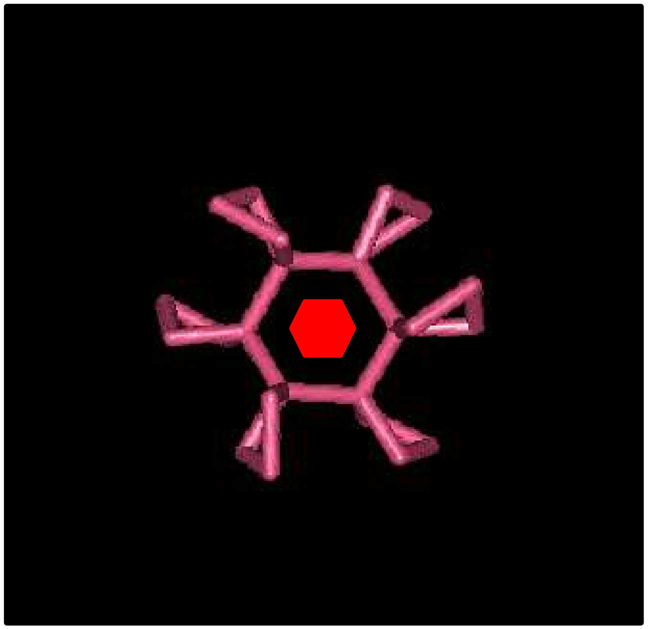


Tungsten oxyfluoride (WOF_4)

Symmetry Elements in 2D



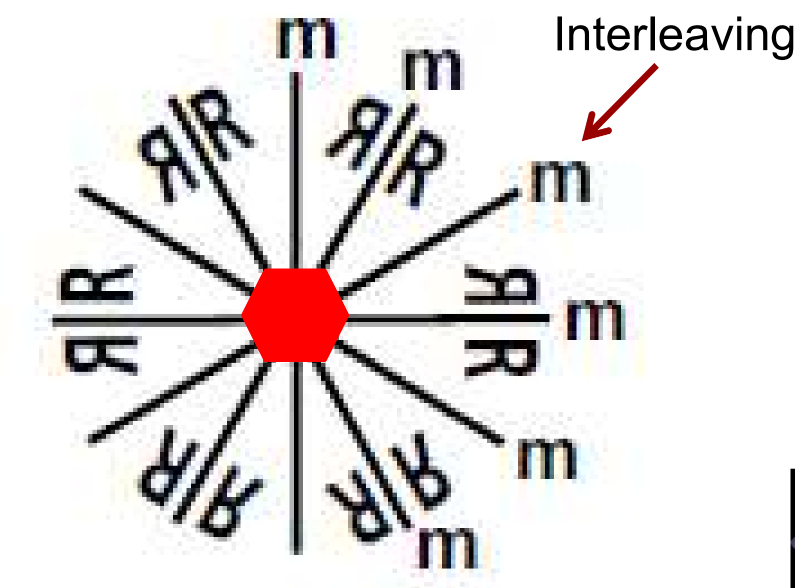


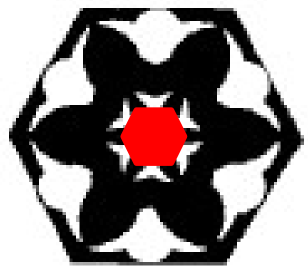


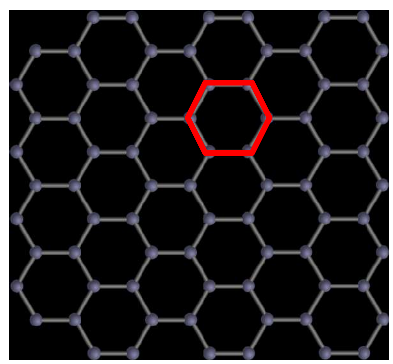
6-rotane ($C_{18}H_{24}$)

Point group: **6**

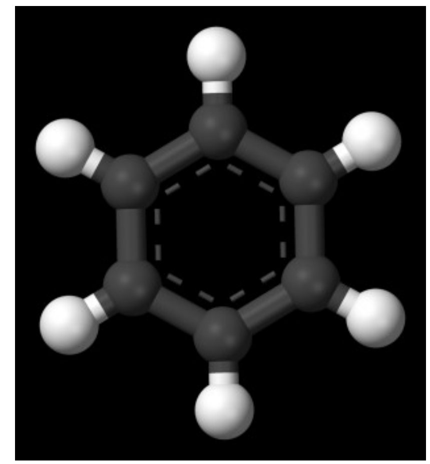
Symmetry Elements in 2D

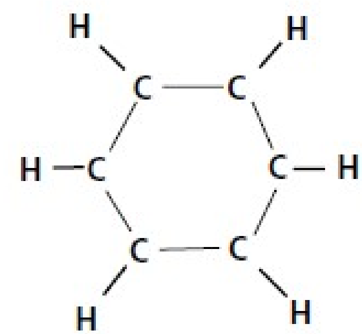






Graphene





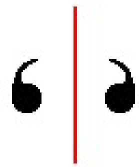
Benzene, C_6H_6

Combination of hexad and mirrors

Point group: **6mm**

Total: 10 Point Groups in 2D

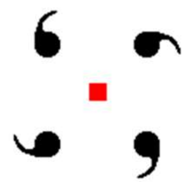
1



m

2

3



4

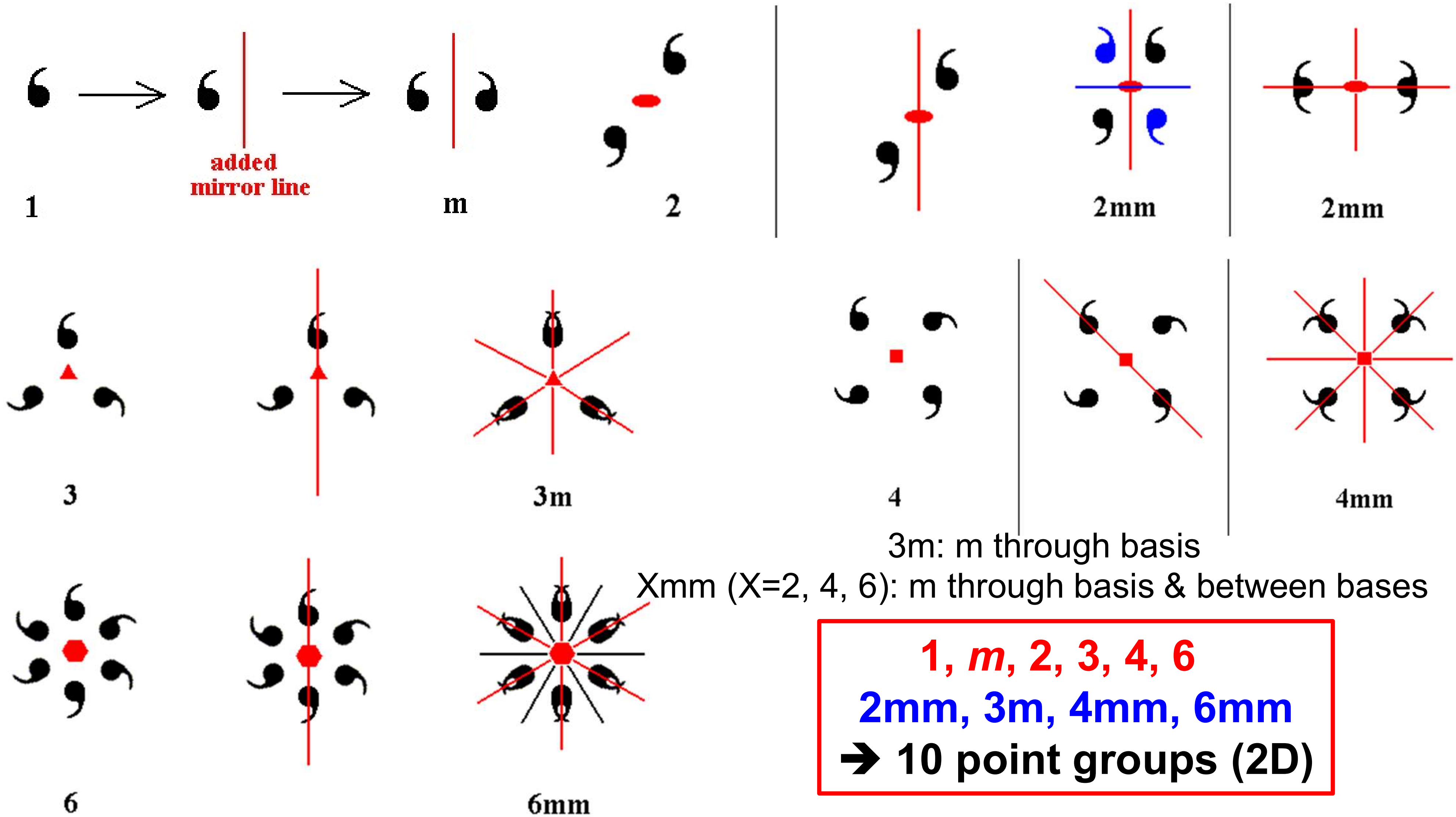
6

1, *m*, 2, 3, 4, 6

Combination of symmetry elements

http://www.metafysica.nl/d2_lattice_1.html

Total: 10 Point Groups in 2D



3m: m through basis

Xmm (X=2, 4, 6): m through basis & between bases

1, *m*, 2, 3, 4, 6

2mm, 3m, 4mm, 6mm

➔ 10 point groups (2D)

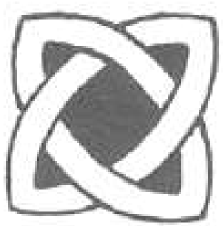
Combination of symmetry elements

http://www.metafysica.nl/d2_lattice_1.html

In-Class Exercise 3

Identify as many symmetry elements as you can and find the point group of the following images.



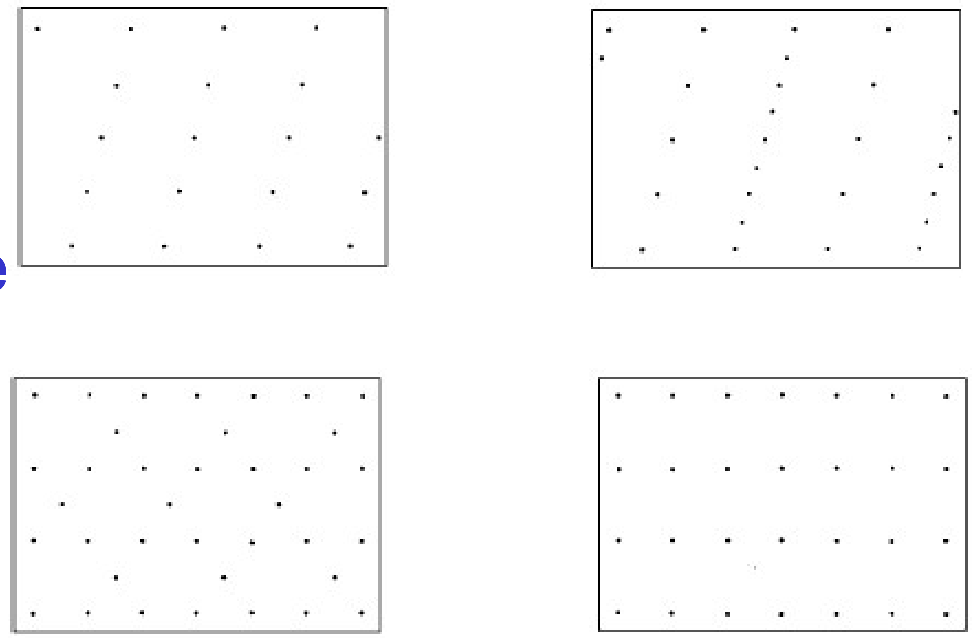




Bravais Lattice in 2D

A **Bravais lattice** is an array of points in space in which the environment of each point is identical.

2D lattice types

oblique

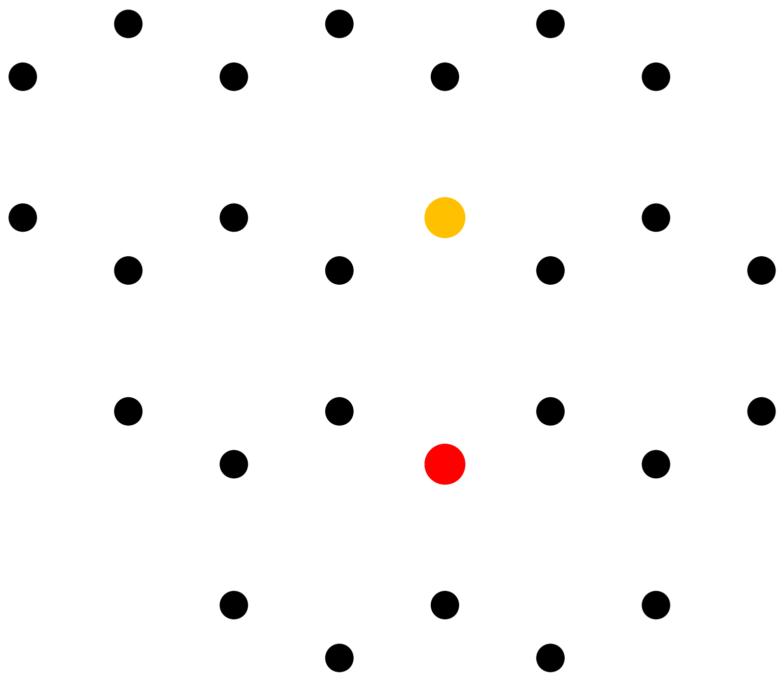


rectangular

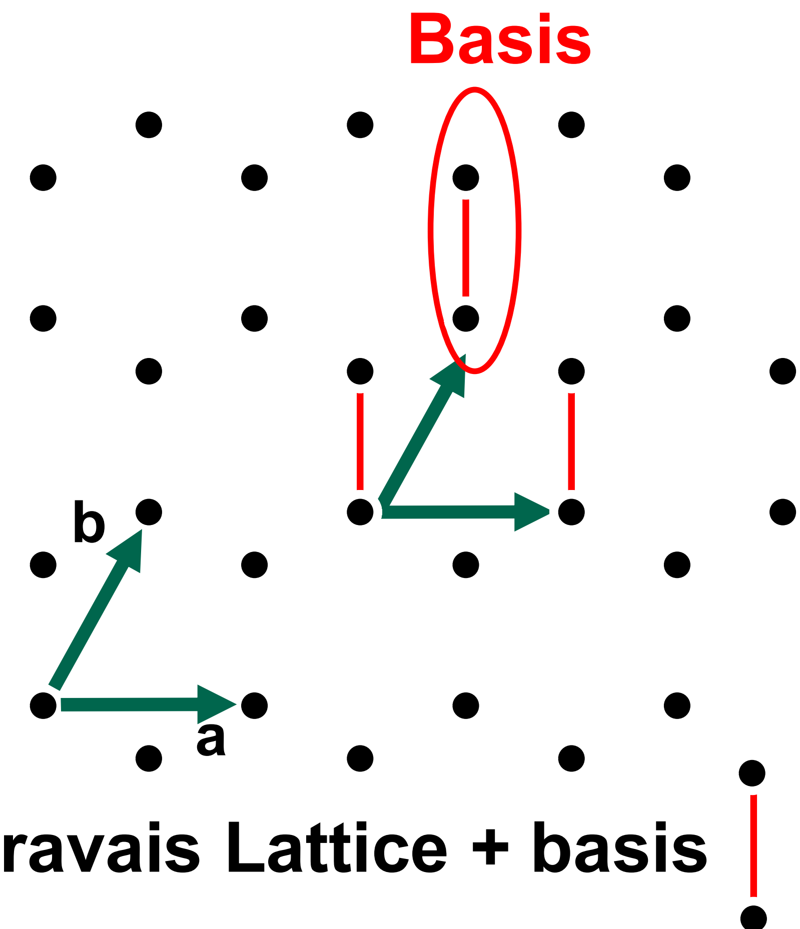
Which arrays constitute a lattice?

- All are regular arrays, but is the surrounding of each point identical?

Bravais lattice in 2D



Not a Bravais Lattice



Bravais Lattice + basis

Bravais lattice: *infinite array of discrete points with an arrangement and orientation that appears exactly the same regardless of the point from which the array is viewed*

All points defined by position vectors of the form $\mathbf{R} = u\mathbf{a} + v\mathbf{b}$

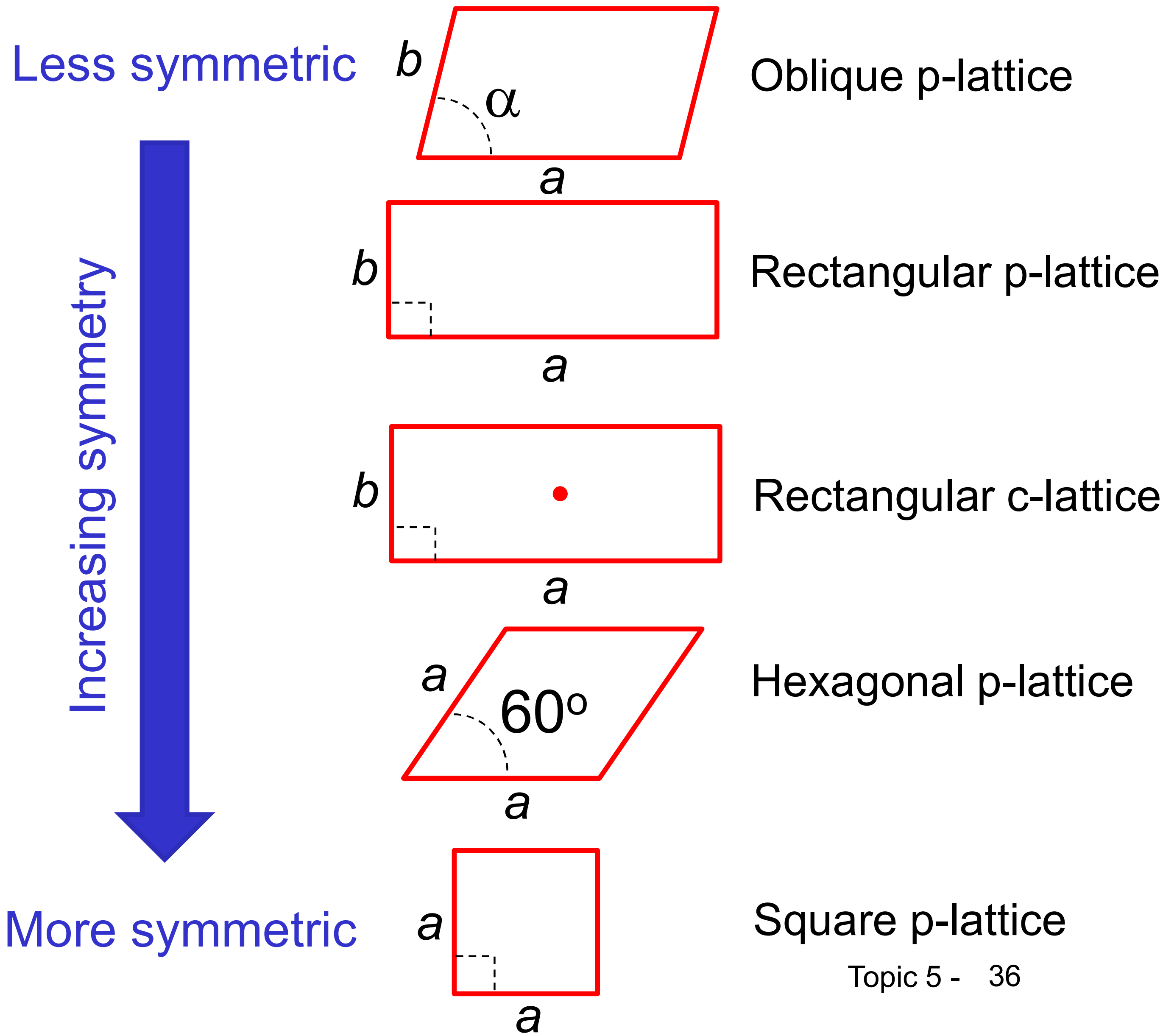
Where *u* and *v* are integers, and **a** and **b** are unit cell vectors.

Five 2D Bravais Lattices

Five plane lattices

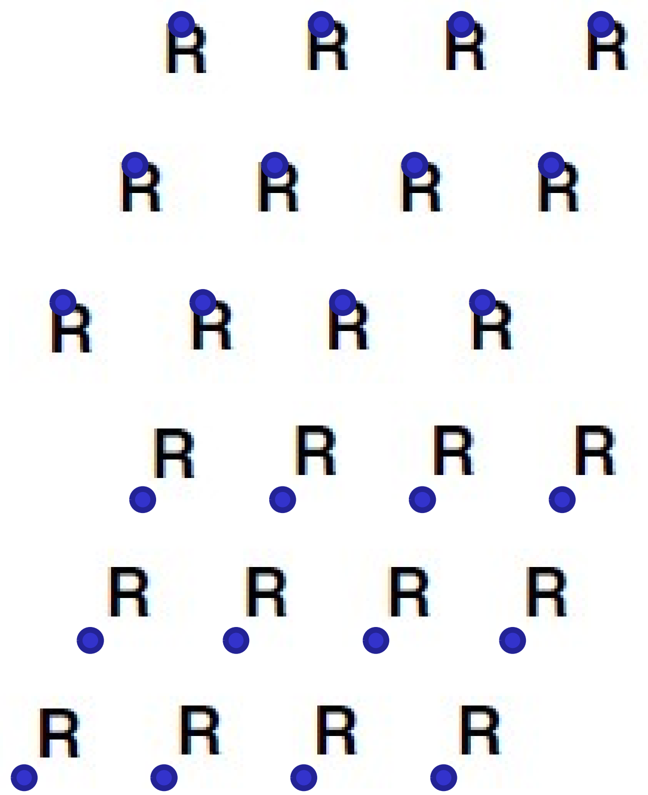
p: primitive
Only one lattice point per unit cell

c: centered
Two lattice points per unit cell

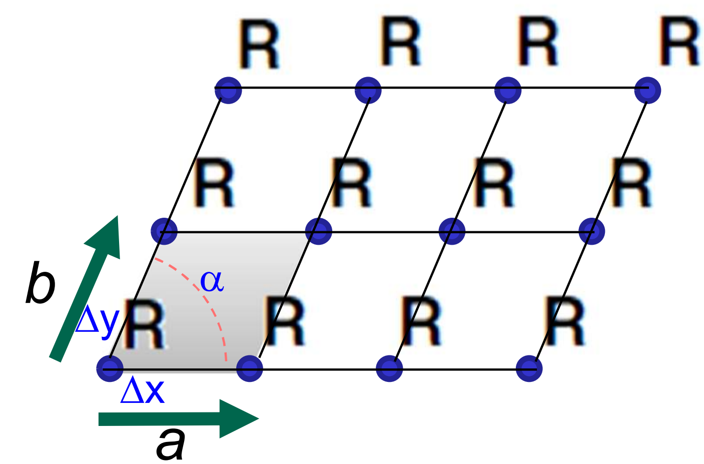


Unit Cell in 2D

Assume an infinite 2D array of **R**; the **R** is a repeating pattern, known as a **basis** on a lattice •



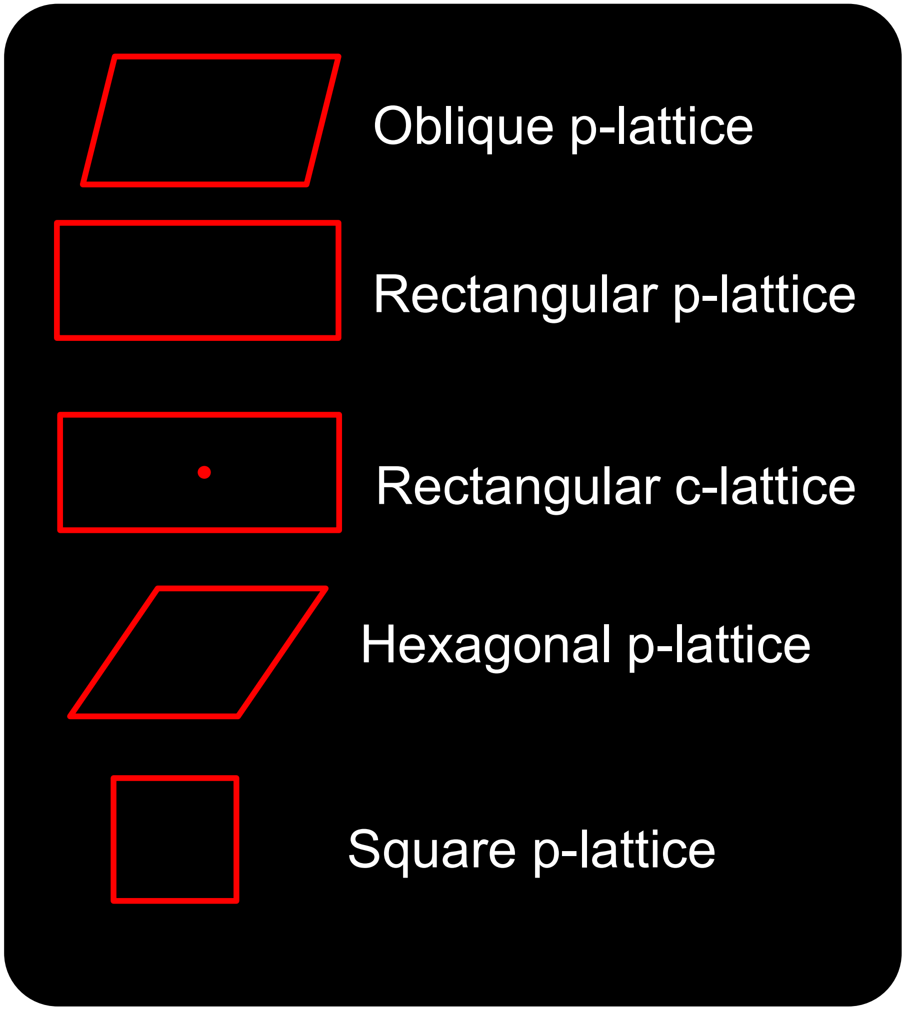
Lattice points: anywhere will do!
Only requirement: same position chosen relative to basis.



If we know the size and shape of the unit cell, the basis and its position/orientation within it, then we can reproduce the structure indefinitely

∴ the unit cell and basis define the entire pattern or structure in 2D

Constructing a 2D Lattice



Combining translation (lattice) with one of ten 2D point group symmetries

+

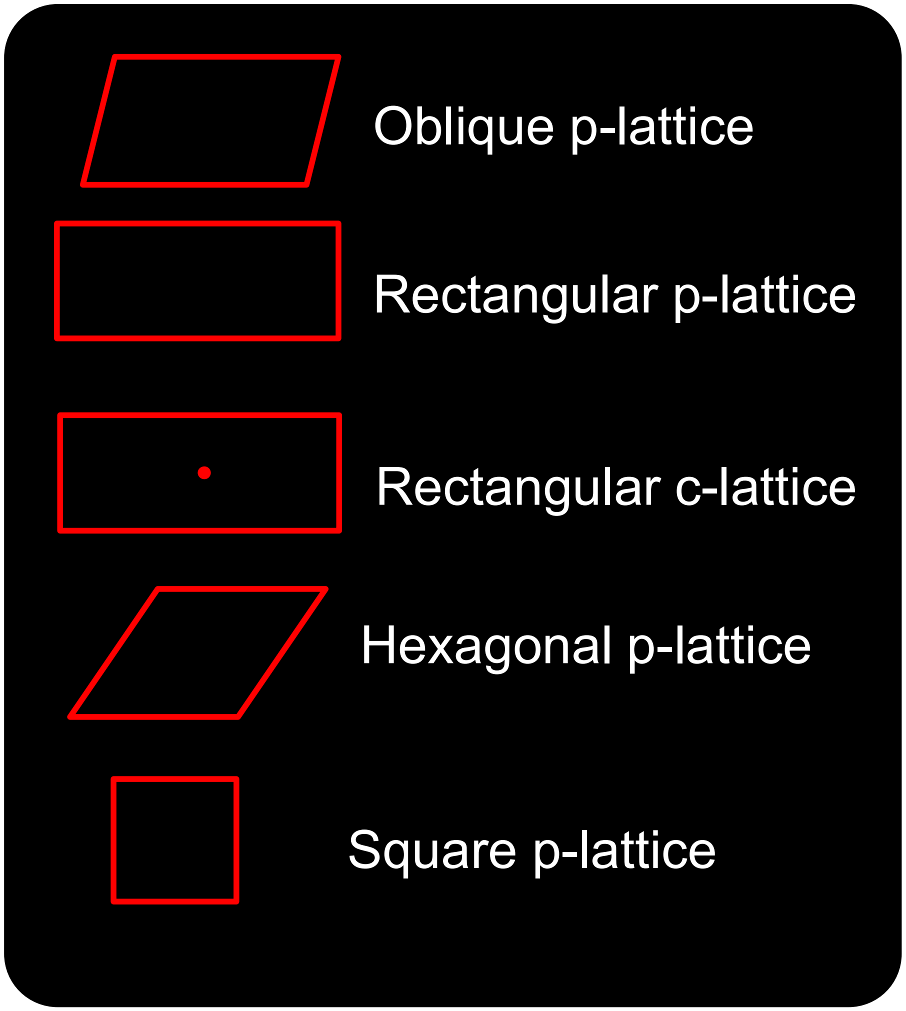
1, *m*, 2, 3, 4, 6
2mm, 3m, 4mm, 6mm

=

Plane groups

A plane group encodes all *essential* symmetries, i.e., symmetries required to replicate the lattice and basis in 2D. Lattice encodes translations.

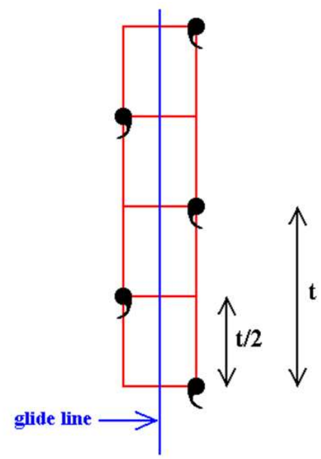
Translation + Point group



+

Combining translation with point group symmetries

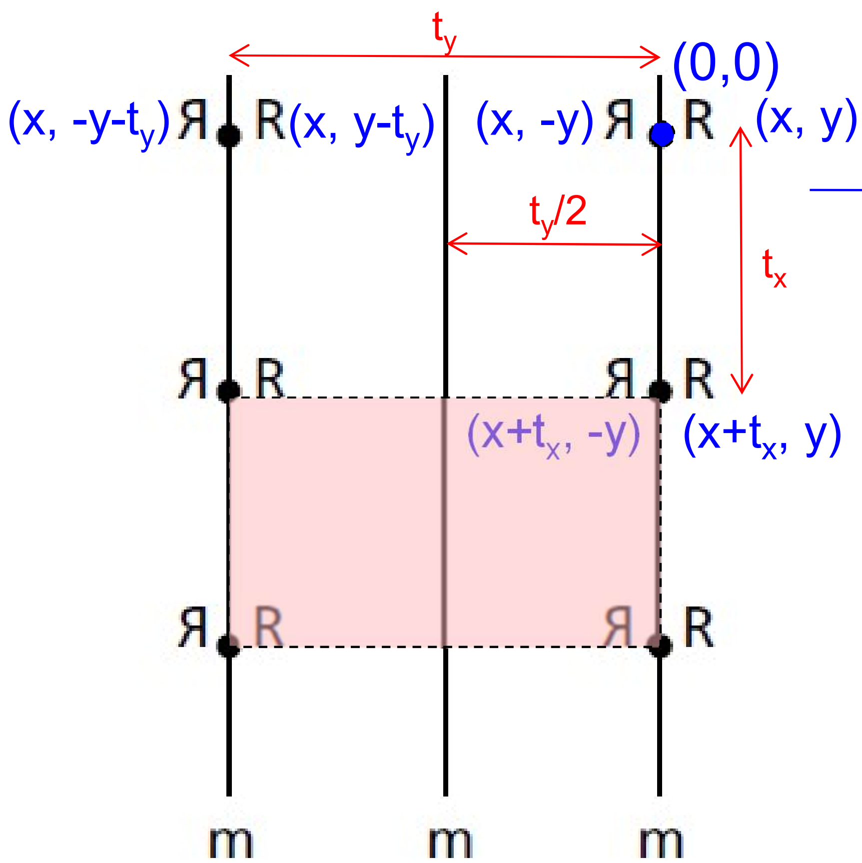
1, *m*, 2, 3, 4, 6
2mm, 3m, 4mm, 6mm

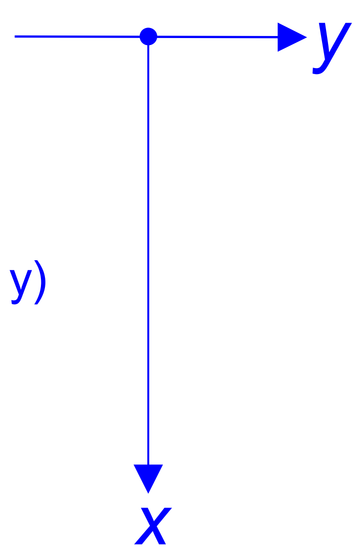


Additional glide line symmetry

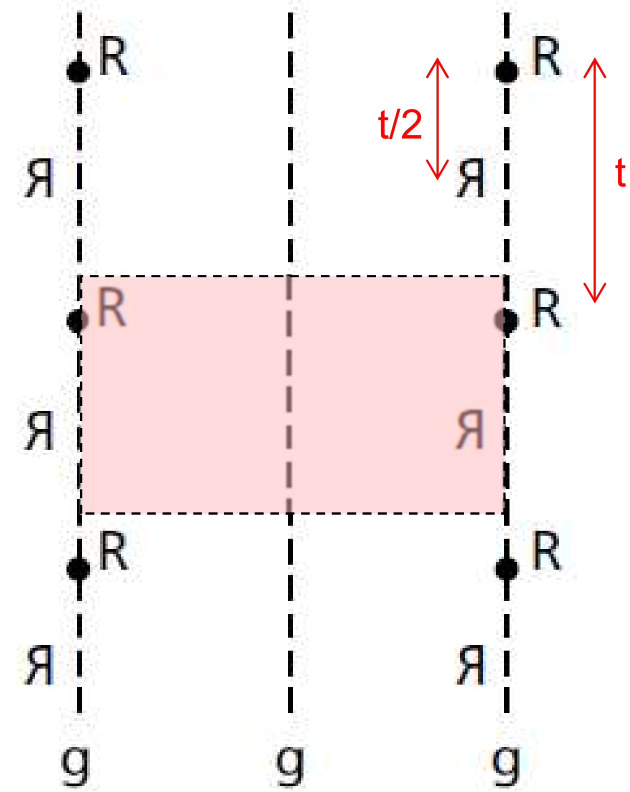
Glide-Reflection Symmetry in a 2D Lattice

Reflection symmetry





Glide-reflection symmetry
a.k.a *reflection glide* or *glide line of symmetry*

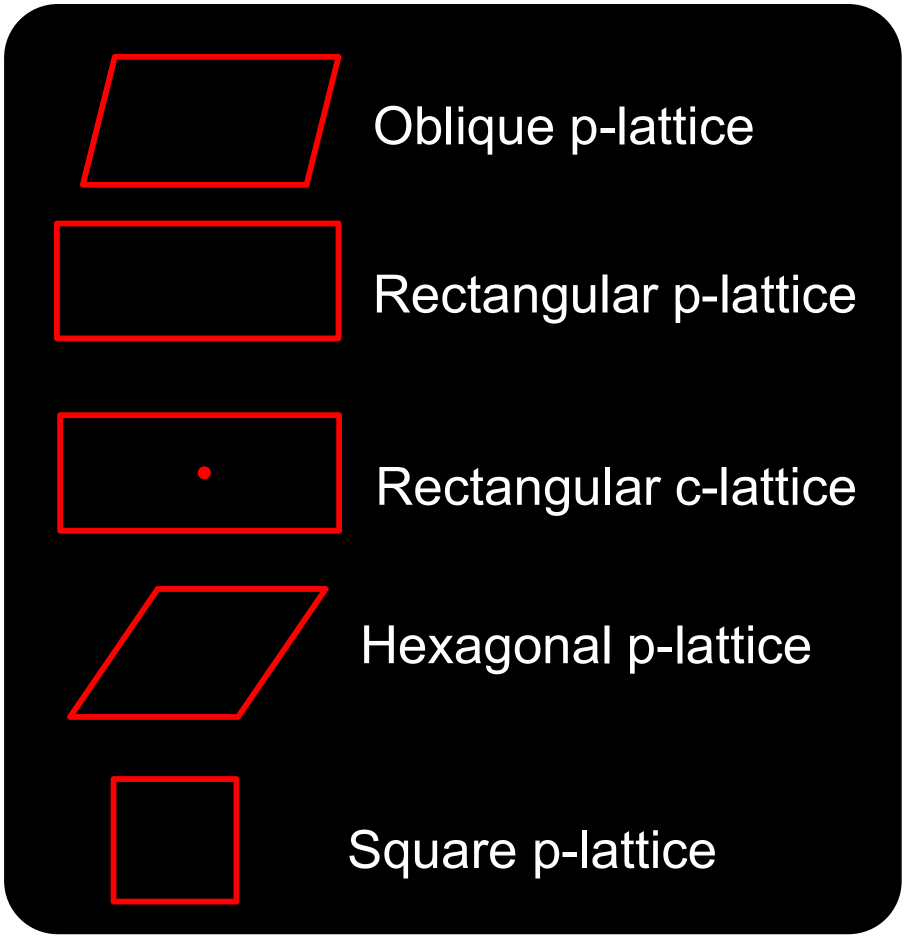


Basis has mirror symmetry

Basis has <u>no</u> mirror symmetry

Glide = translation + reflection
New Plane groups: pg, p2gg, p2mg, p4gm

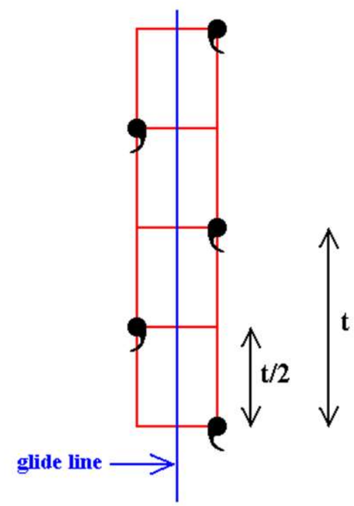
Total: 17 Plane Groups



+

Combining lattice translation with point group symmetries & their combinations

1, *m*, 2, 3, 4, 6
2mm, 3m, 4mm, 6mm



Additional glide line symmetry

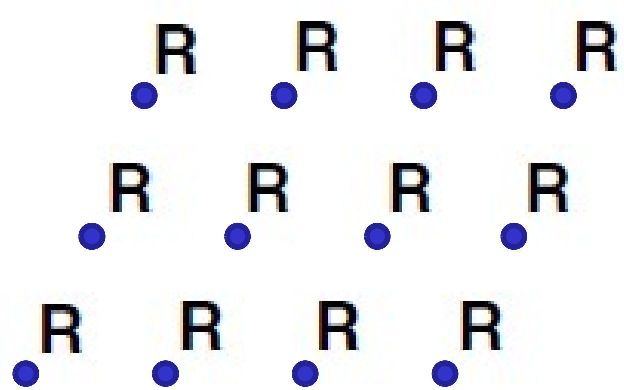
17 distinct plane groups from 5 plane lattices =

p1, pm, p2, p3, p4, p6, pg, cm,
p2mm, p2mg, p2gg, c2mm, p3m1, p31m,
p4mm, p4gm, p6mm

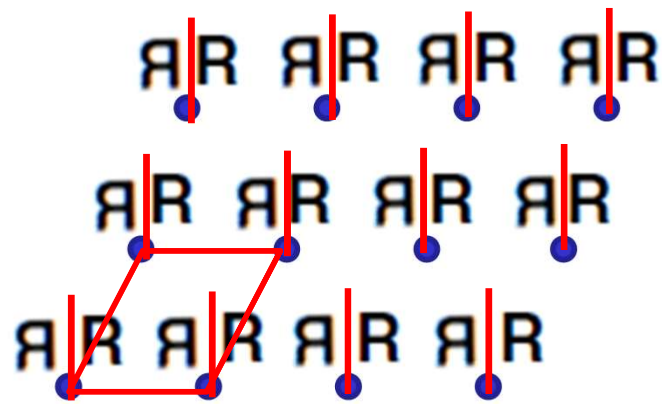
Basis vs. Lattice symmetry

R

Asymmetric basis



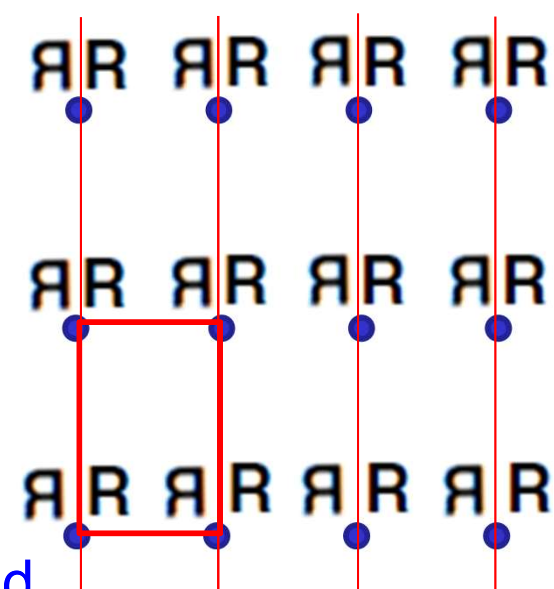
Oblique: asymmetric lattice



oblique

p1

Vertical mirror line of symmetry within basis is not compatible with oblique lattice!



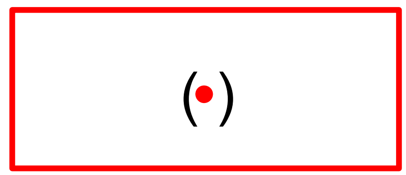
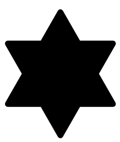
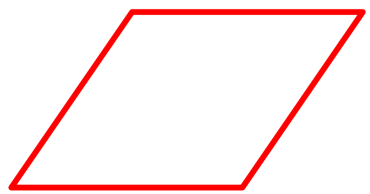
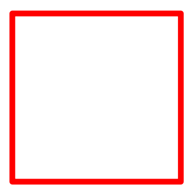
rectangular

pm

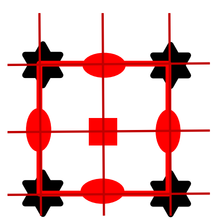
Symmetry must be present both in lattice and in basis to also be present in plane group

Preserving Basis Symmetry in a Lattice

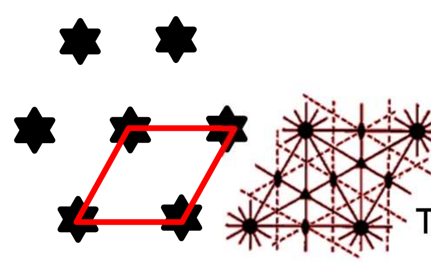
Basis symmetry and the least symmetric lattice required to preserve symmetry of basis in extended lattice

Basis	Symmetries?	Lattice?	
R	No symmetries		Oblique p-lattice
M	1 mirror		Rectangular p(c)-lattice
	1 hexad		Hexagonal p-lattice
+	1 tetrad + 2 mirrors		Square p-lattice

Examples:

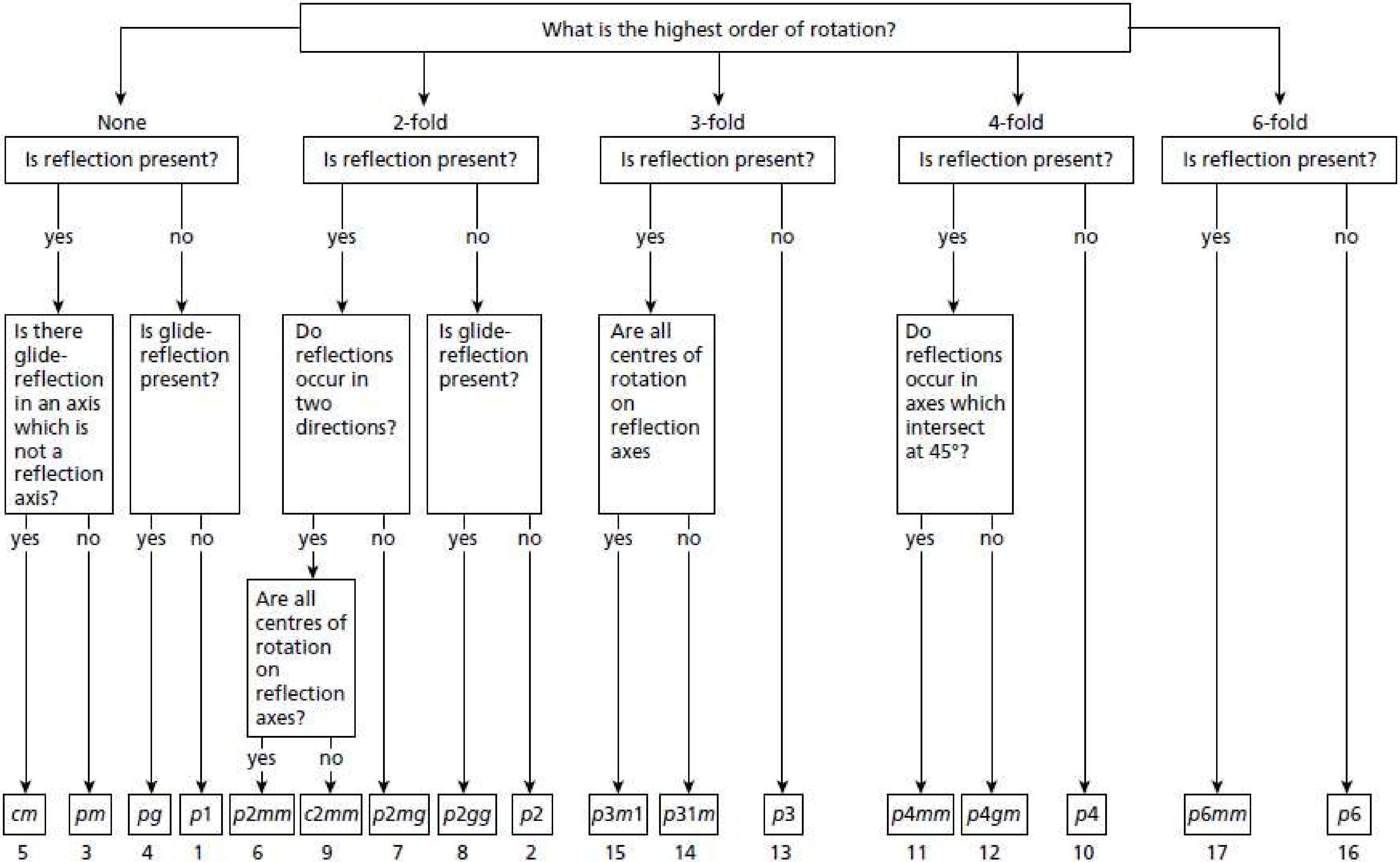


Hexad of basis is lost in extended lattice due to square lattice



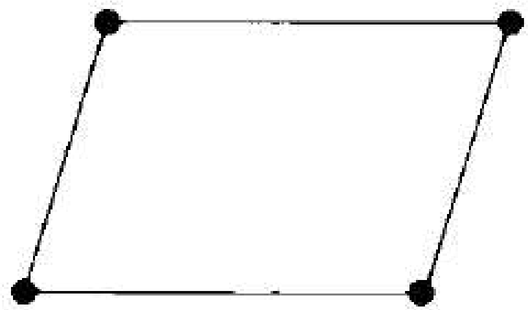
Hexad of basis preserved in extended lattice due to hexagonal lattice

Plane Groups Flowchart



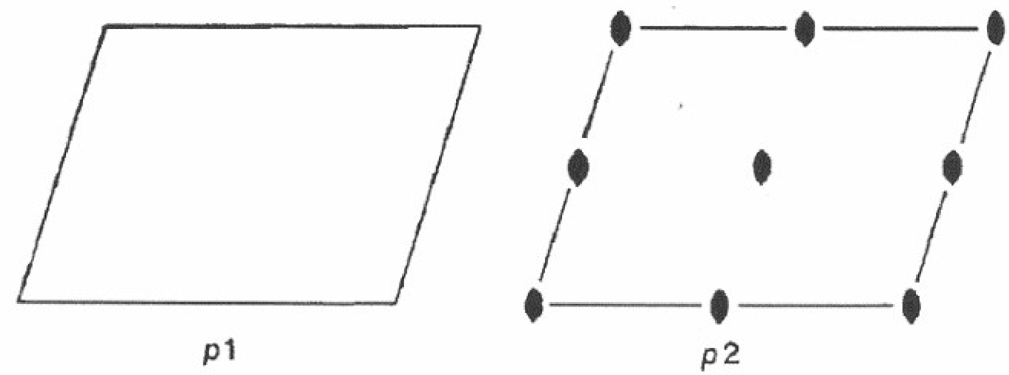
Oblique Plane Groups

Bravais lattice

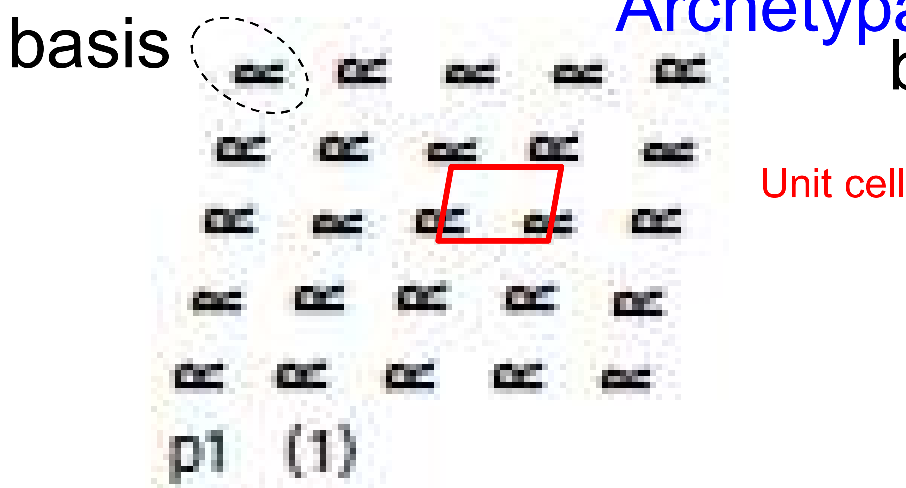


The oblique *p*-lattice

Plane groups



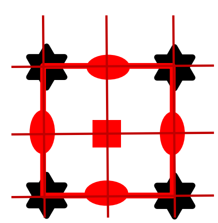
Archetypal lattice + basis



2 oblique plane groups

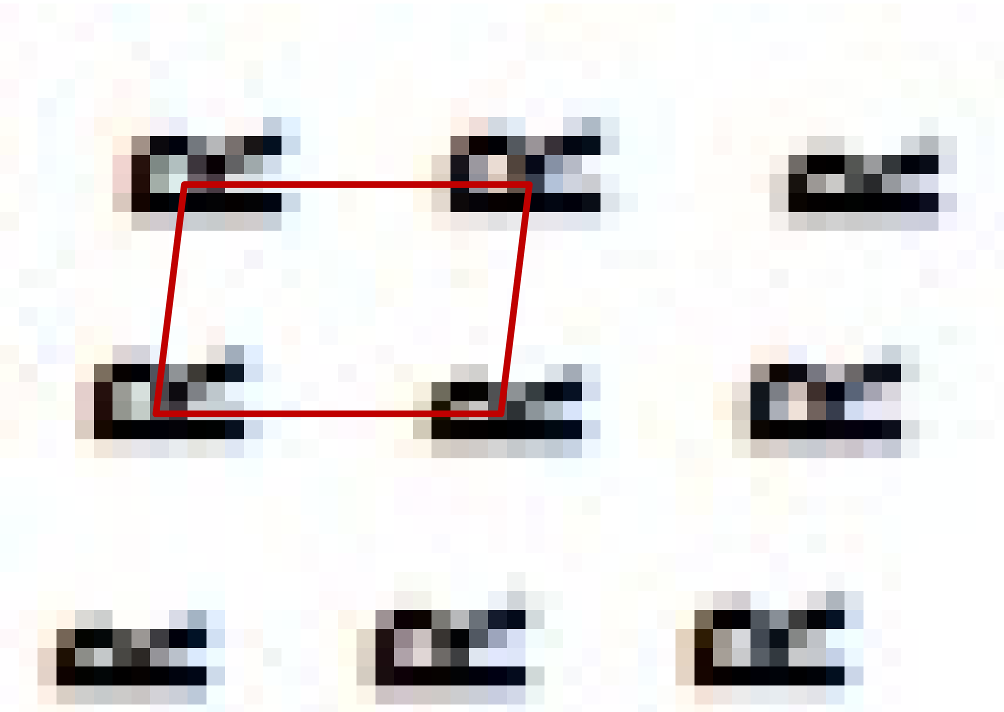
Oblique lattice can only sustain monad and diad as highest symmetries ➔ p1 & p2

Note: A higher symmetry basis can be present, but its symmetry does not carry to the lattice.

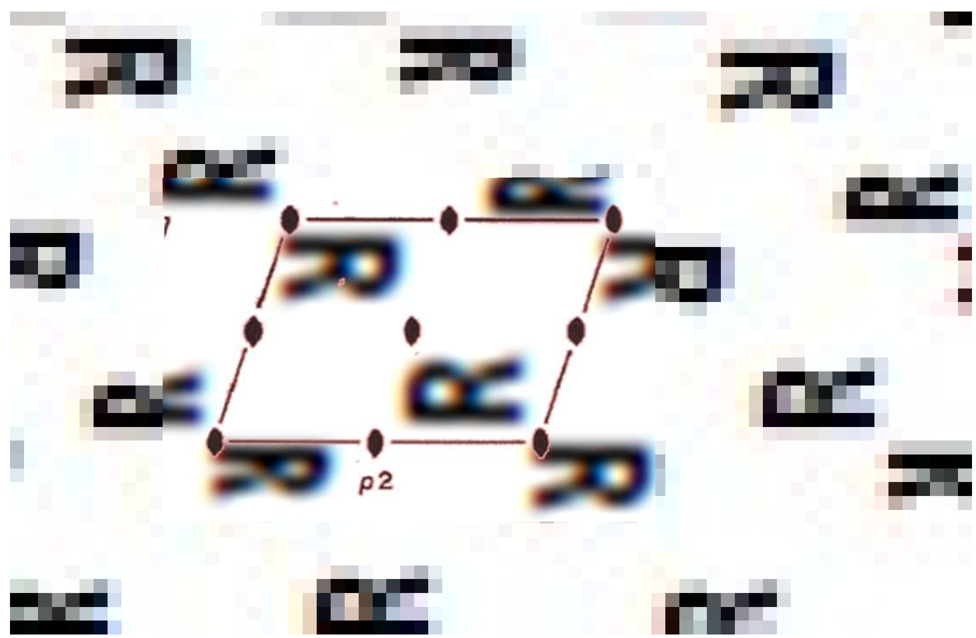


e.g., basis containing hexad on a square lattice

Oblique Plane Groups



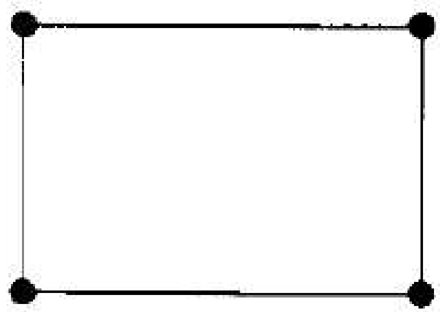
p1



p2

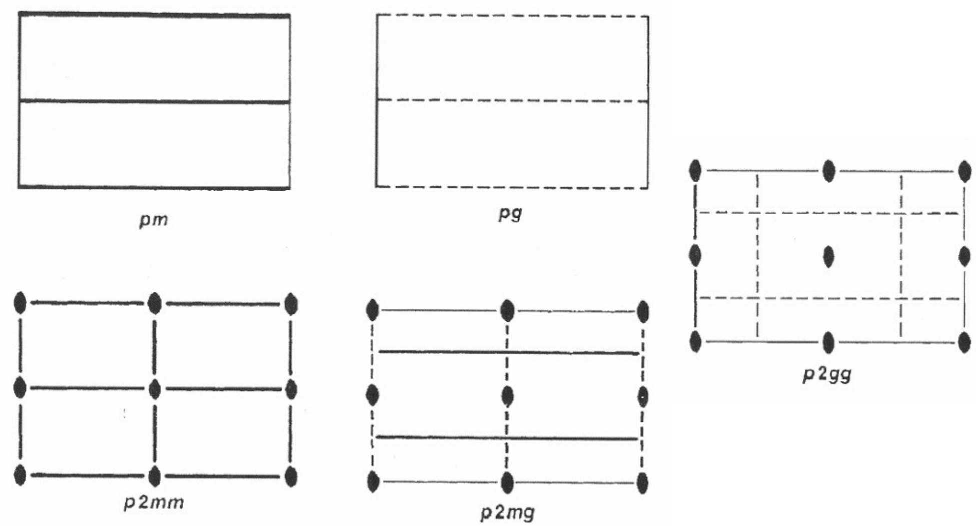
Rectangular p Plane Groups

Bravais lattice

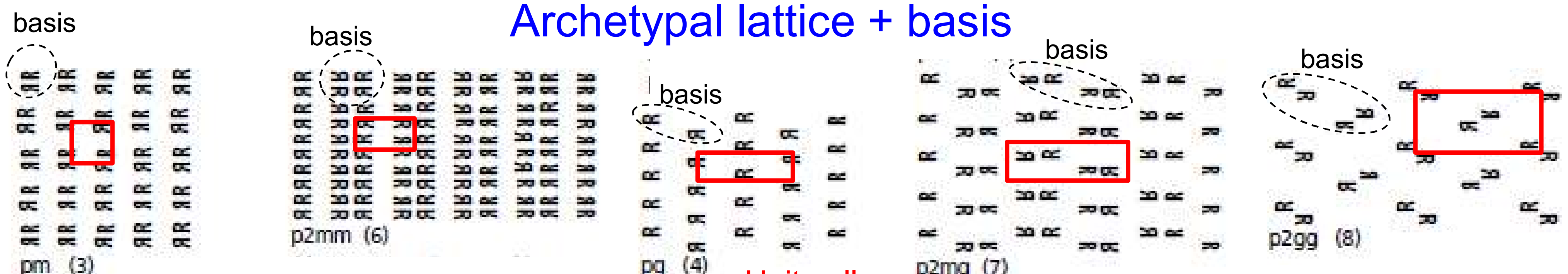


The rectangular *p*-lattice

Plane groups



Archetypal lattice + basis

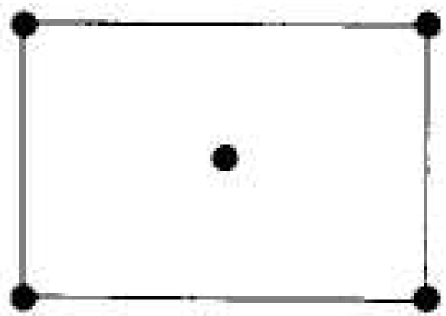


Rectangular p lattice is compatible with diad, mirror & glide ➔ pm, pg, p2mm, p2mg, p2gg

5 rectangular p plane groups

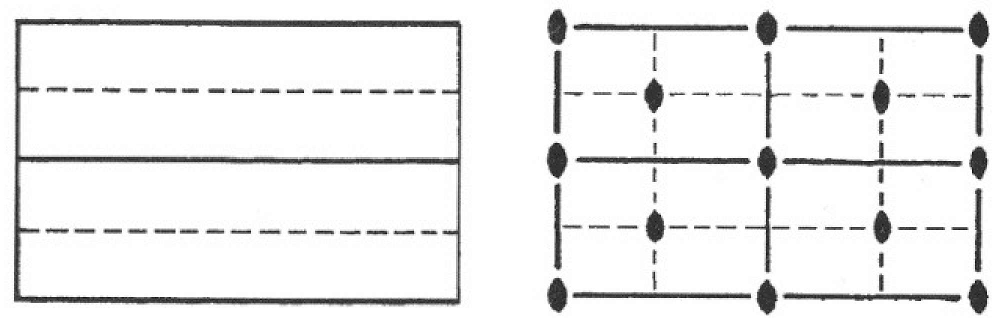
Rectangular c Plane Groups

Bravais lattice



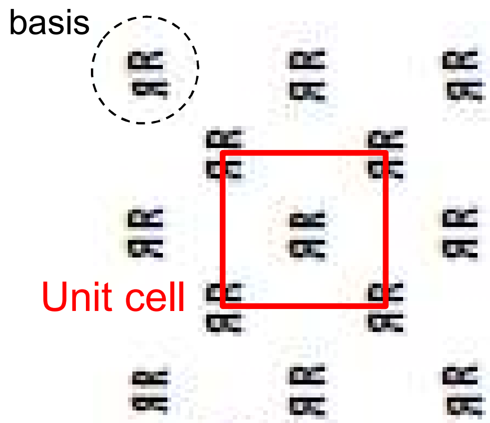
The rectangular *c*-lattice

Plane groups

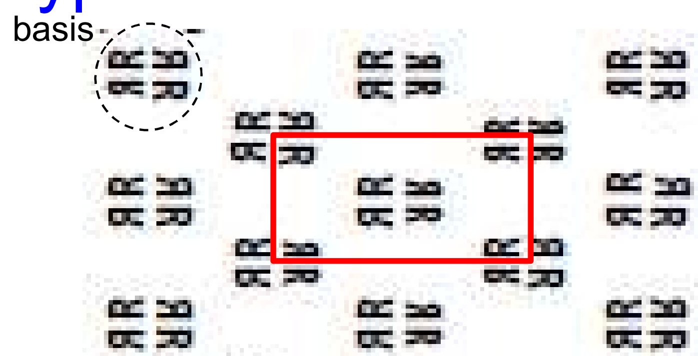


cm *c2mm*

Archetypal lattice + basis



cm (5)



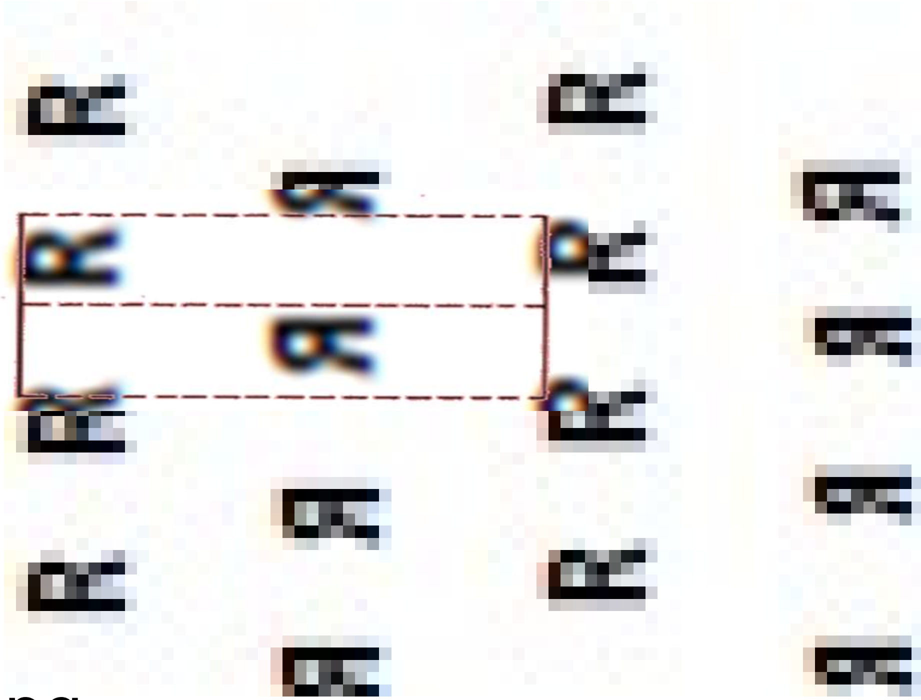
c2mm (9)

Basis appears twice per unit cell since it is rectangular **c** instead of **p**.

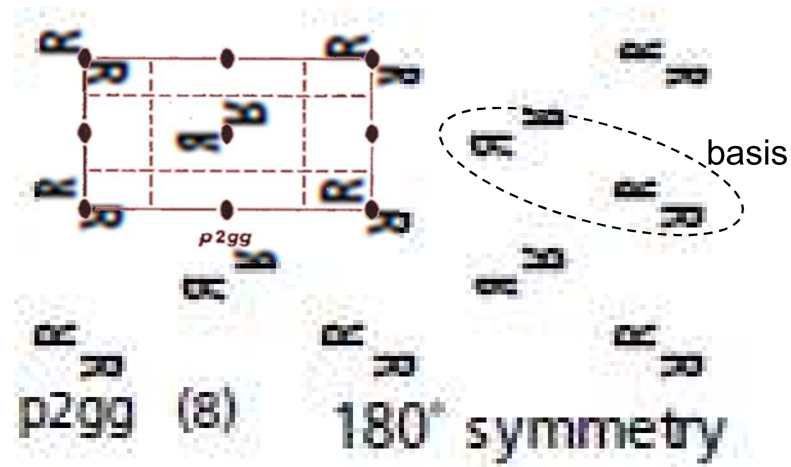
2 rectangular c plane groups

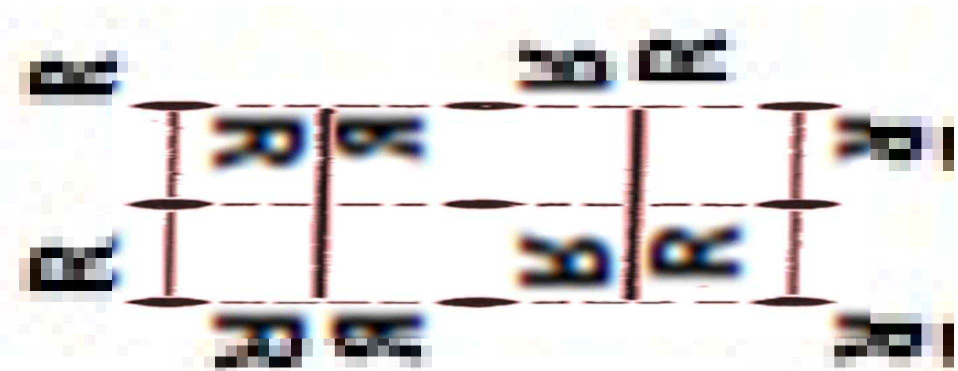
Rectangular c lattice is compatible with diad, mirror & glide ➔ cm, c2mm

Rectangular p & c Plane Groups

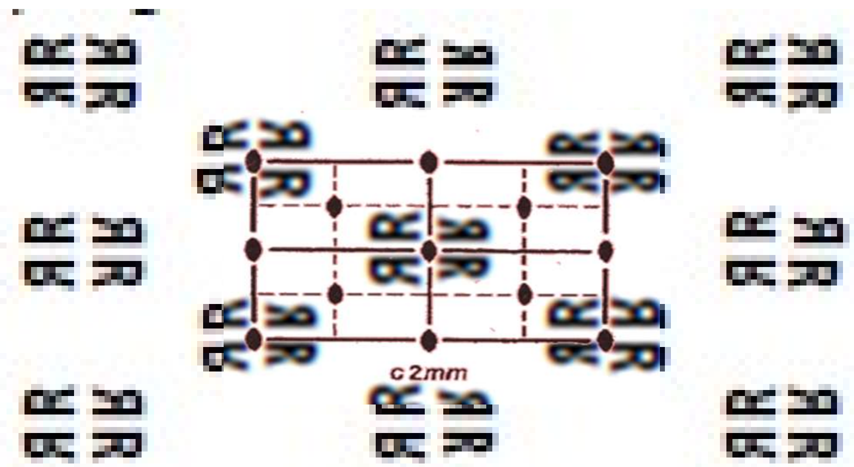


pg



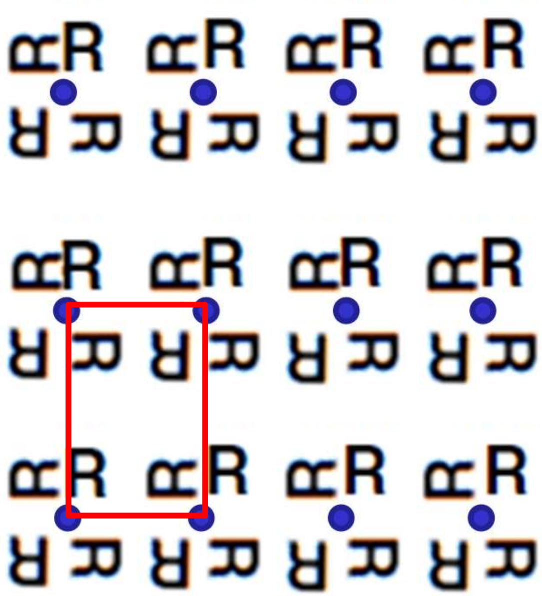


p2mg



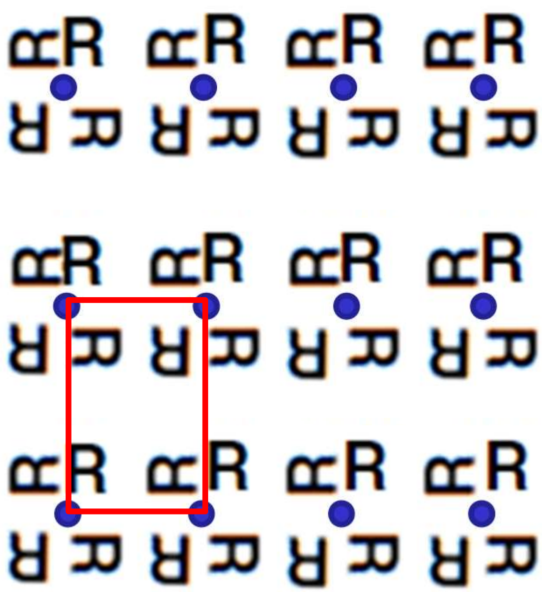
c2mm

Rectangular vs Square Lattice

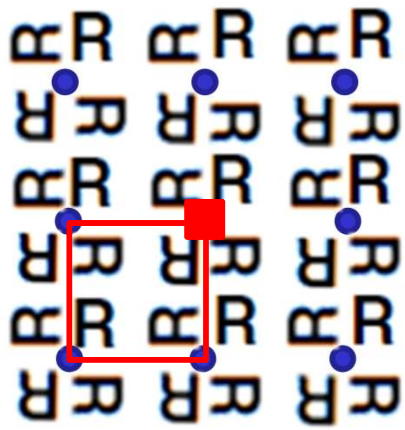


- **Rectangular lattice** only allows 2-fold rotation symmetry (**diad**)

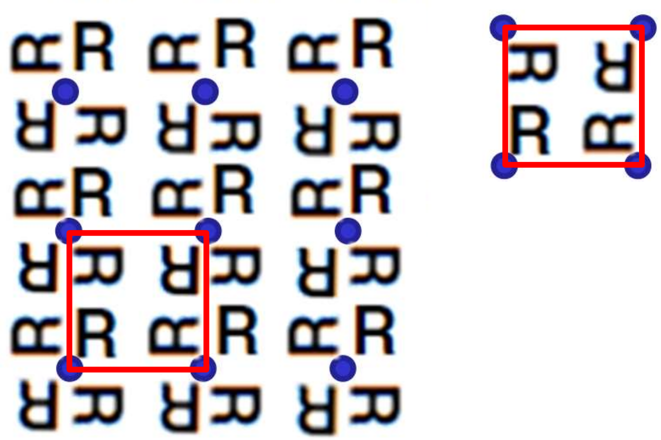
Rectangular vs Square Lattice

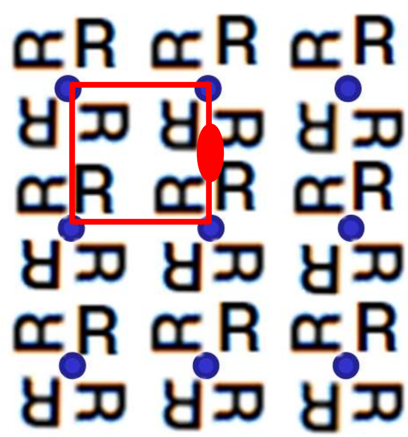


- **Rectangular lattice** only allows 2-fold rotation symmetry (**diad**)
- **Square lattice** is required for 4-fold rotation symmetry (**tetrad**)



4-fold rotation symmetry (**center & corner**)

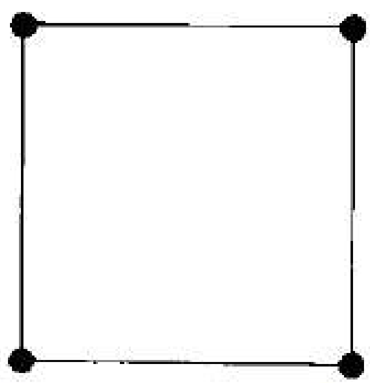




2-fold rotation symmetry (**edge**) (implied)

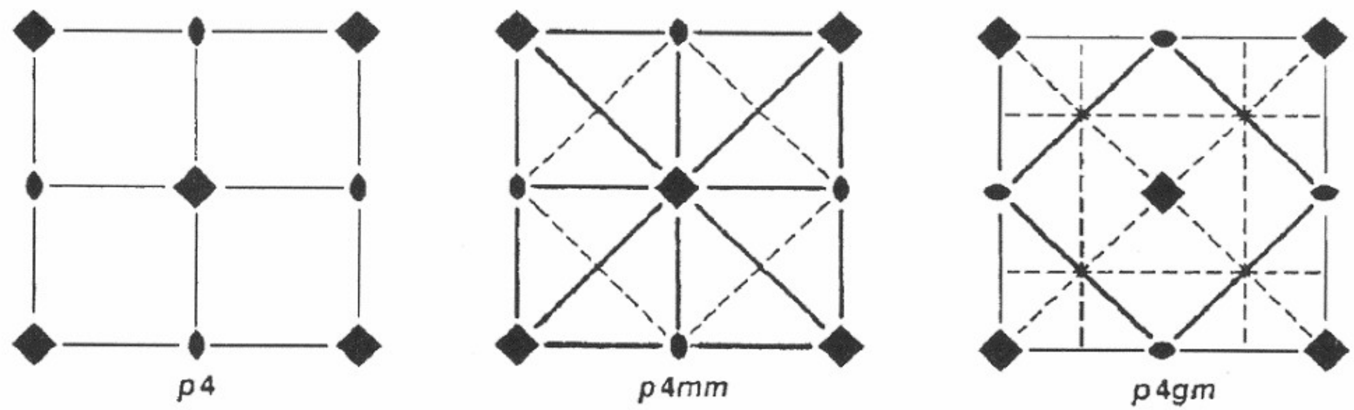
Square Plane Groups

Bravais lattice

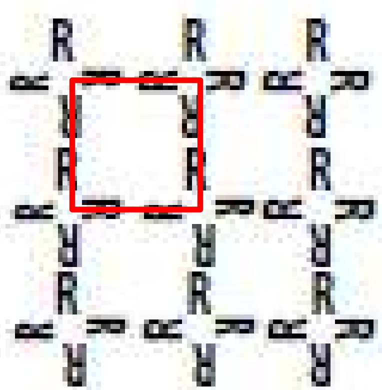


The square p-lattice

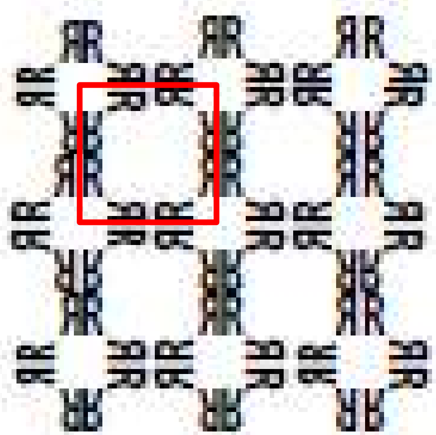
Plane groups



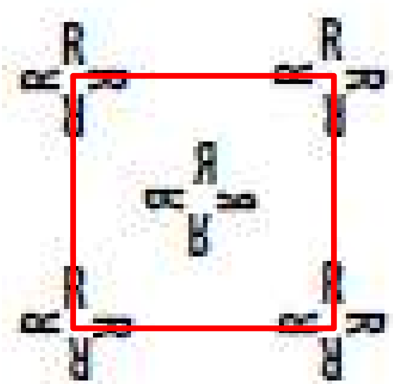
Archetypal lattice + basis



p4 (10)



p4mm (11)

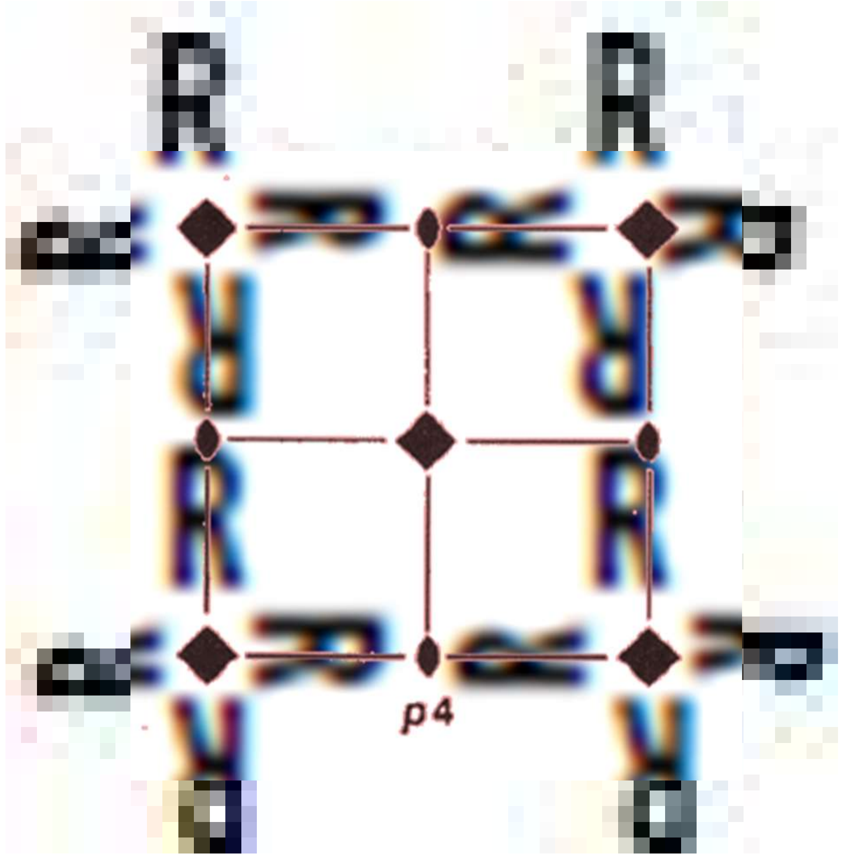


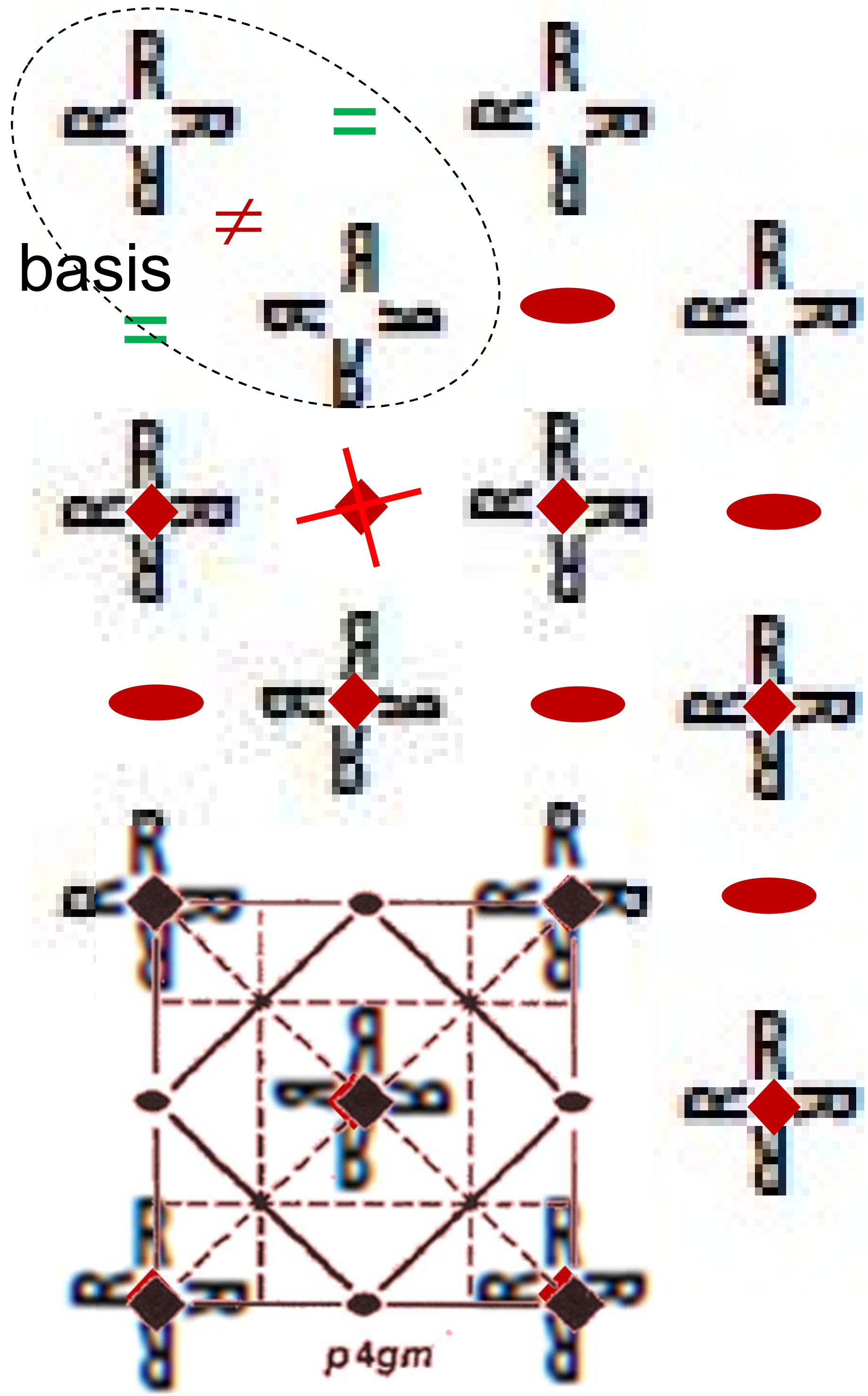
p4gm (12)

3 square plane groups

Square lattice is compatible with tetrad, diad, mirror & glide ➔ p4, p4mm, p4gm

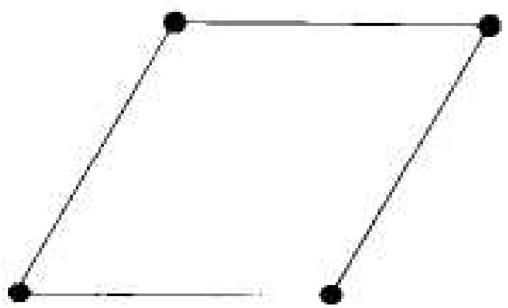
Square Plane Groups





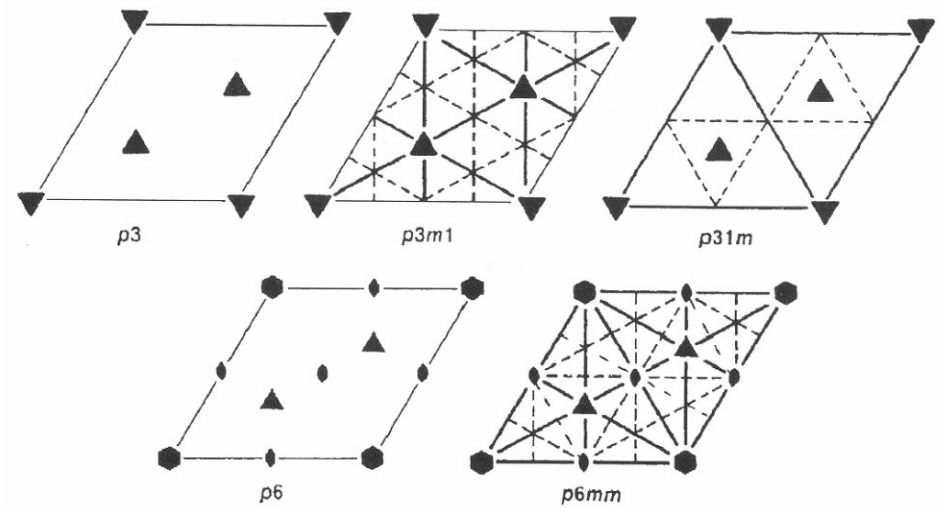
Hexagonal Plane Groups

Bravais lattice

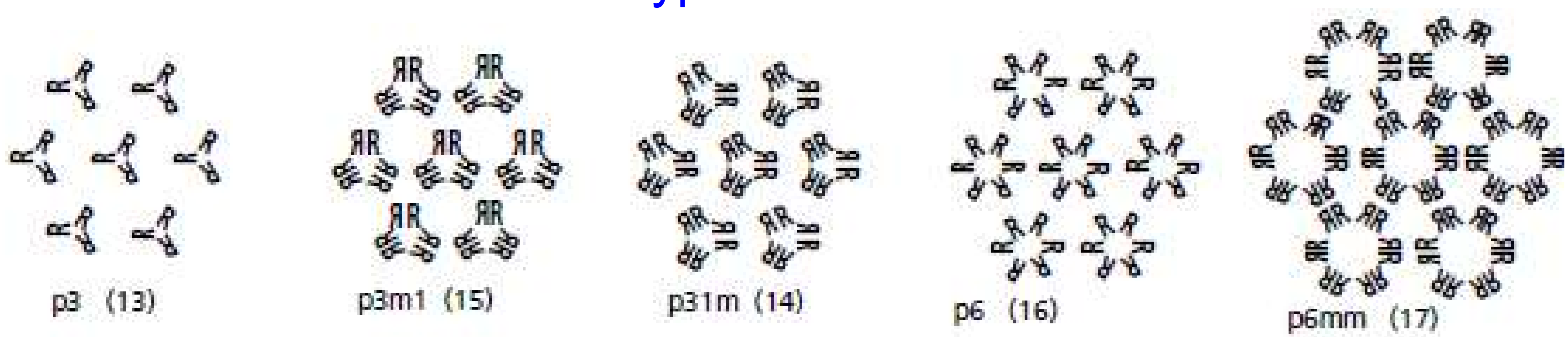


The hexagonal *p*-lattice

Plane groups



Archetypal lattice + basis



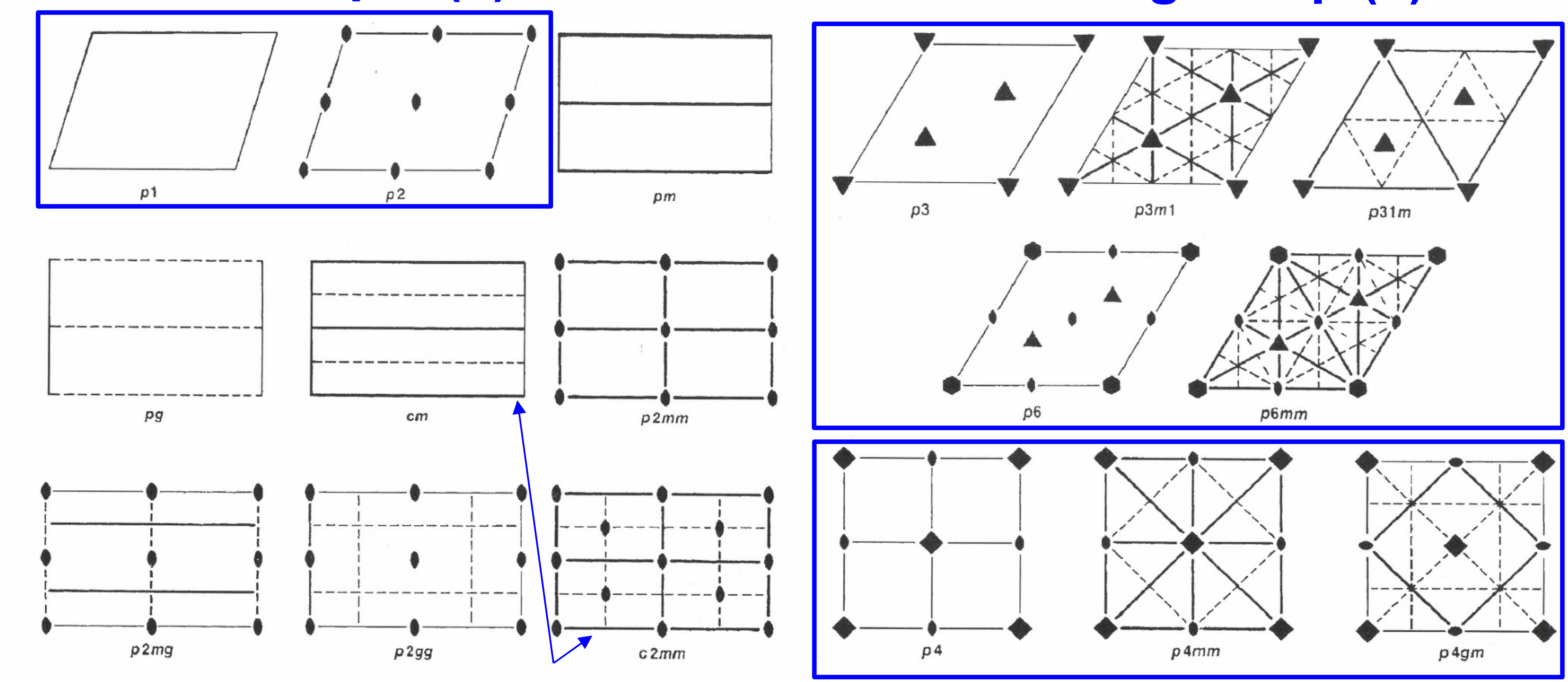
Hexagonal lattice compatible with hexad, triad, mirror & glide ➔ p3, p3m1, p31m, p6, p6mm

5 hexagonal plane groups

Total: 17 Plane Groups

Oblique (2)

Hexagonal p (5)



rectangular p (5) **rectangular c (2)**

Square p (3)

Decoding Plane Groups

p1, pm, p2, p3, p4, p6, pg, cm, p2mm, p2mg, p2gg, c2mm, p3m1, p31m, p4mm, p4gm, p6mm

- Which 2D Bravais lattice does a plane group correspond to?
- Highest rotational symmetry?
- Plane group provides essential symmetry elements needed to replicate a basis over the Bravais lattice

Decoding lattice from the plane group:

1st position: c → **rectangular c**

2nd position: *m* or *g* alone → **rectangular p**

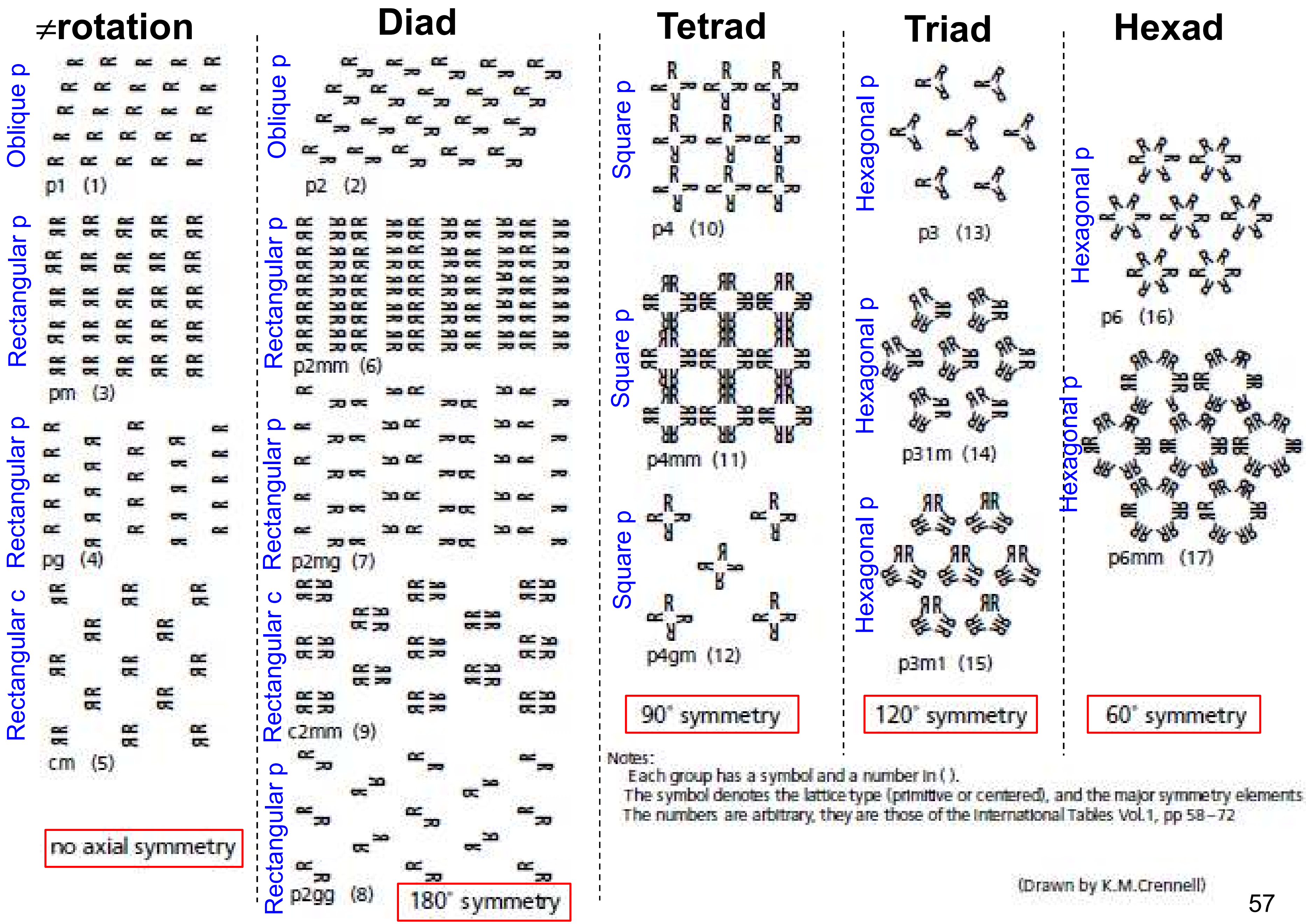
"1" or "2" w/o *m* or *g* → **oblique**

"2" w/ *m* and/or *g* → **rectangular**

"3" or "6" → **hexagonal**

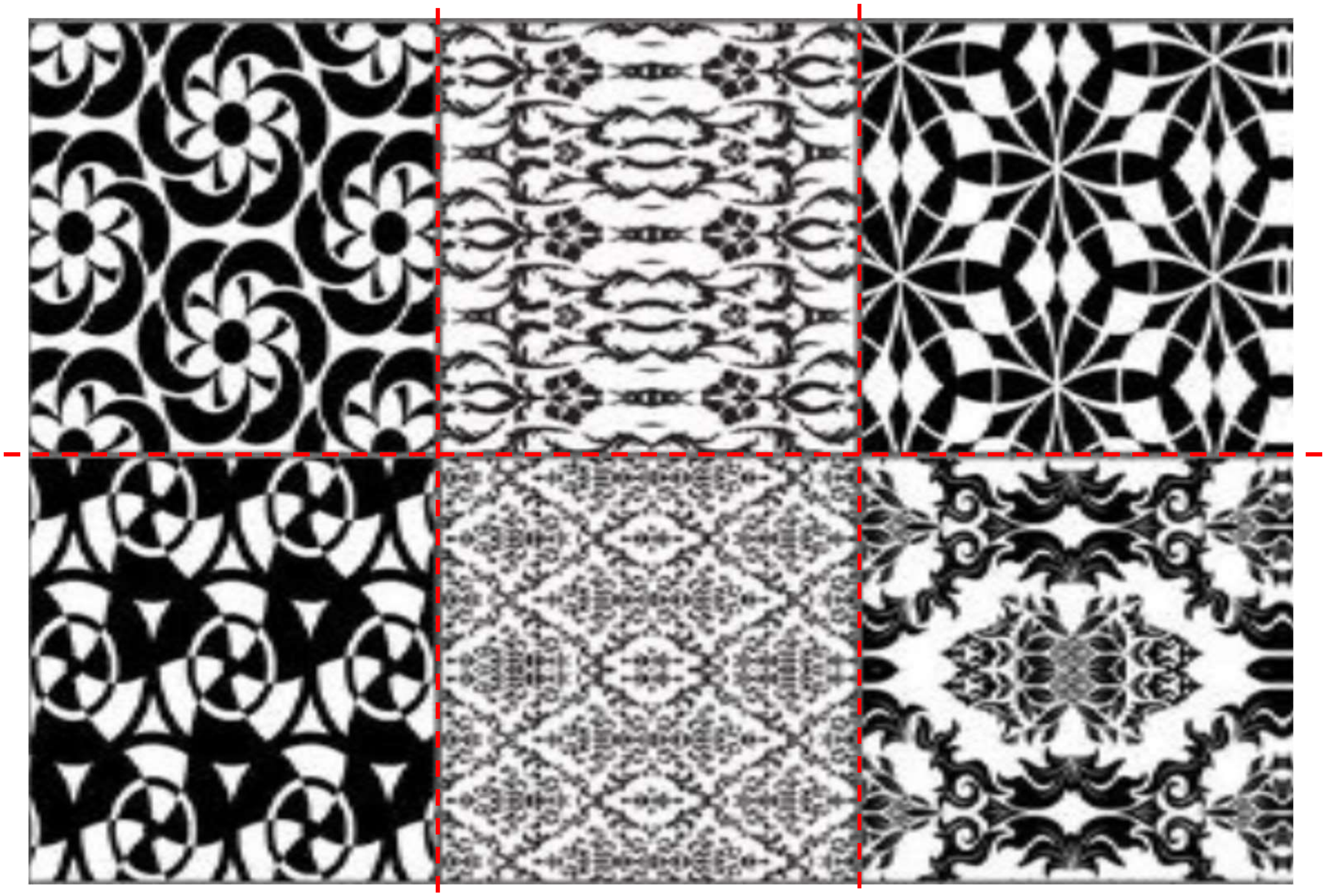
"4" → **square**

Plane Groups by Rotational Symmetry



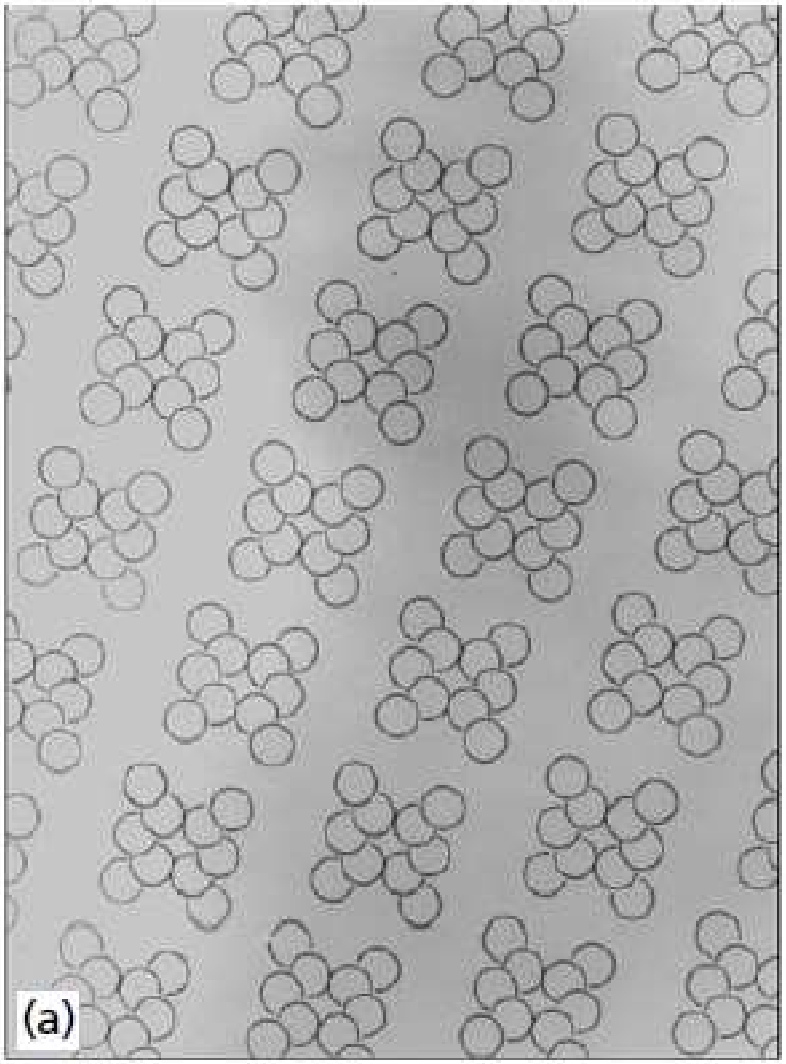
In-Class Exercise 4

Identify the unit cell, as many symmetry elements as you can and the point group of the following structures.



In-Class Exercise 5

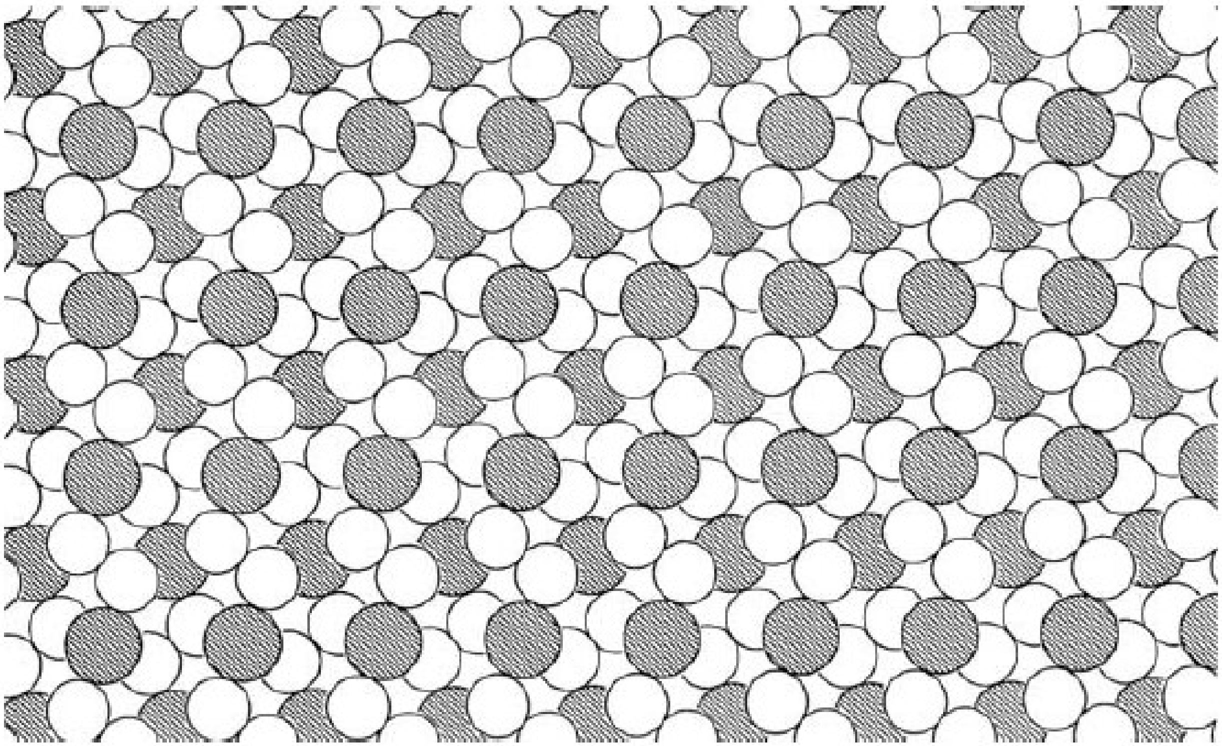
Recognizing unit cells & plane groups



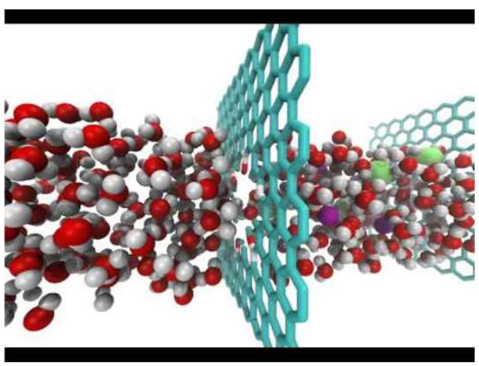
(1)Identify basis
(2)Identify lattice (translation)
(3)Identify unit cell
(4)Identify rotation
(5)Identify mirror

In-Class Exercise 6

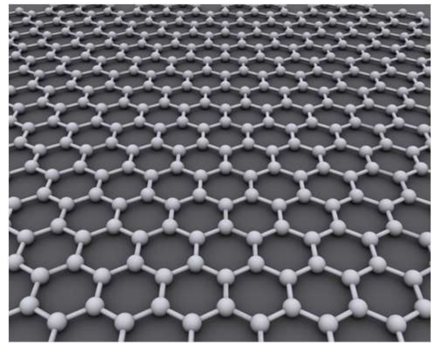
Recognizing unit cells & plane groups
FeS_2, Marcasite

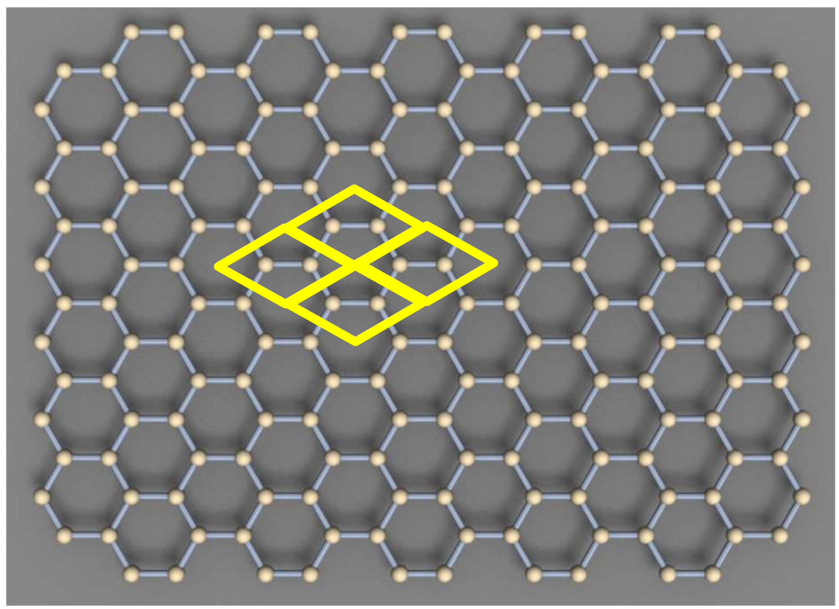


2D Nanomaterial: Graphene

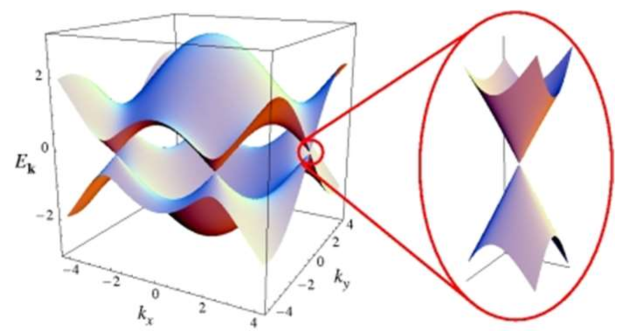


https://www.youtube.com
/watch?v=EOVduH-iGMw





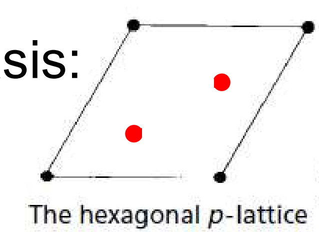
We learnt about unit cells, lattice points and bases.
We learnt about symmetry elements, point groups in 2D and plane groups.

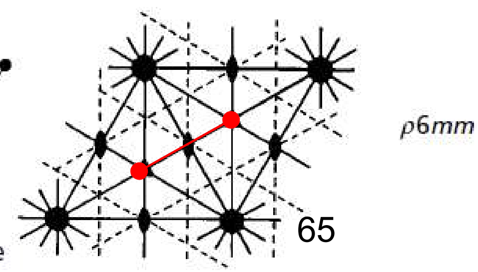


http://condensedconcepts.blogspot.com/2013/04/the-dirac-cone-in-graphene-is-emergent.html

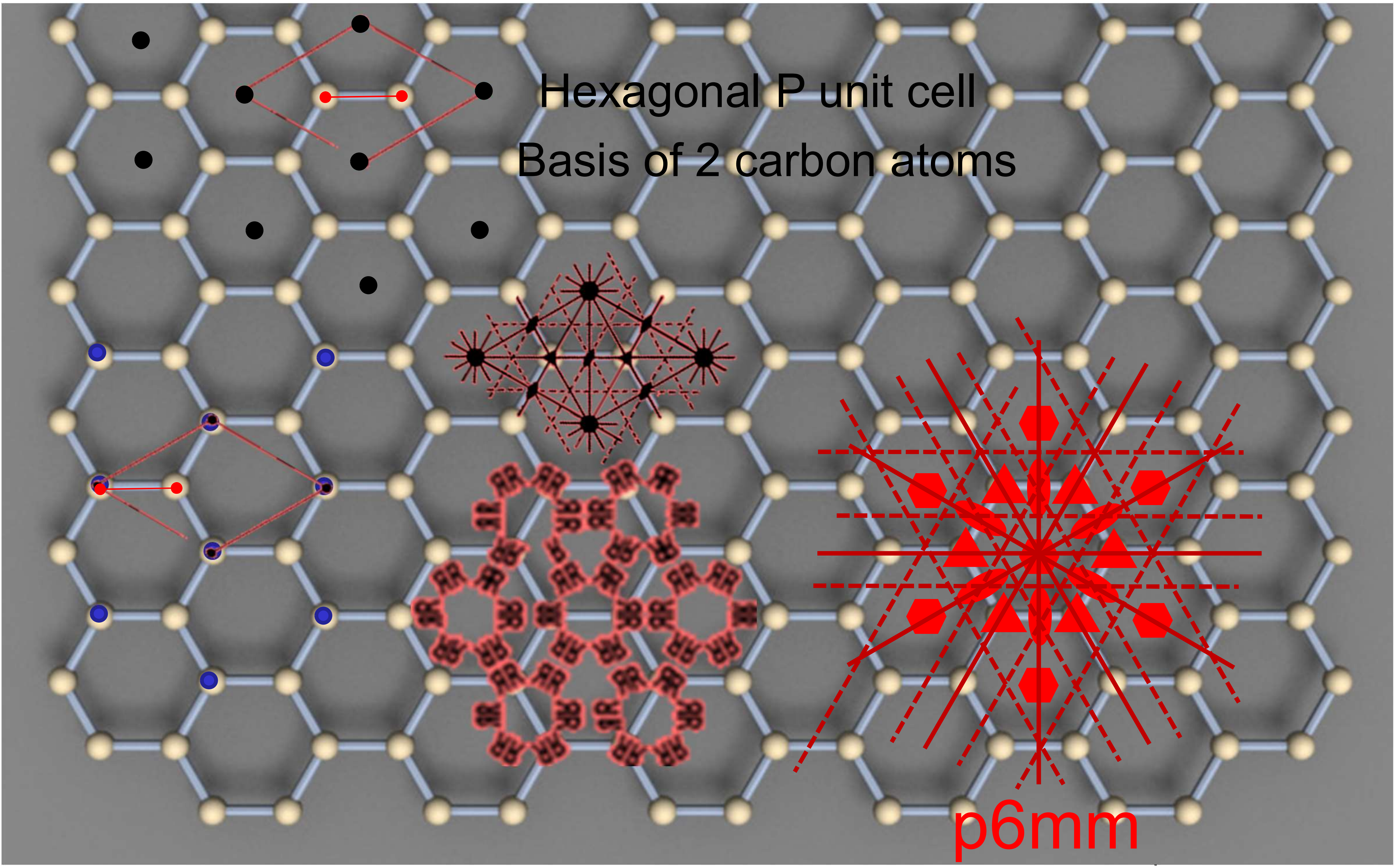
Plane group: p6mm

Basis:

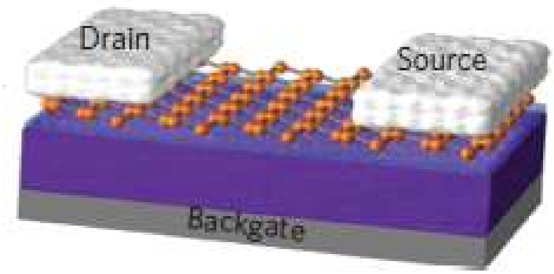




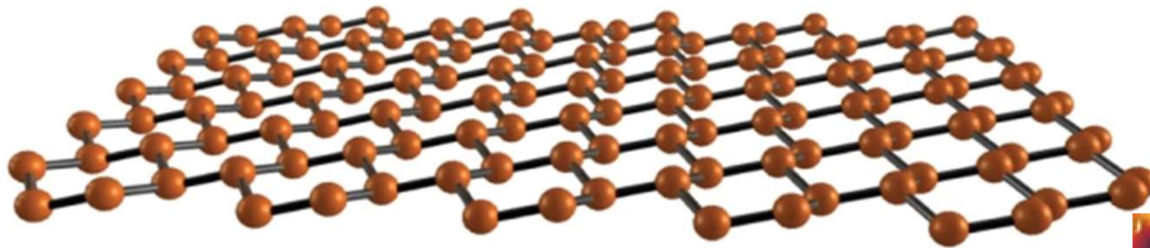
2D Nanomaterial: Graphene

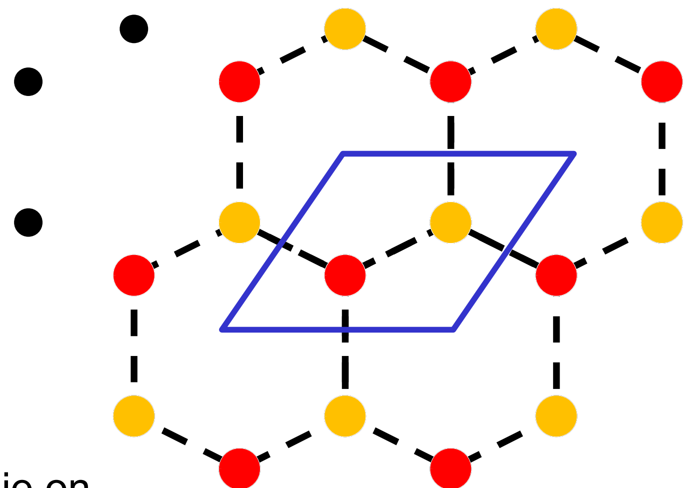


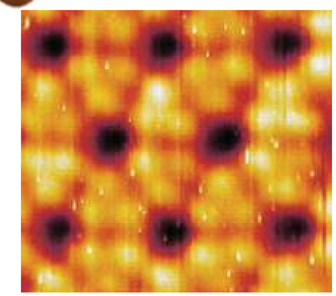
2D Nanomaterial: Silicene

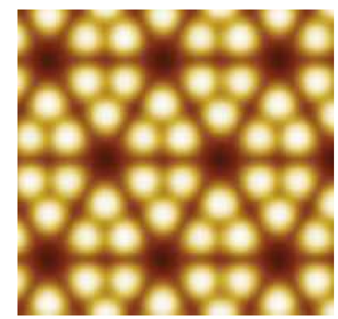


Nature Nano. **10**, 227–231 (2015)







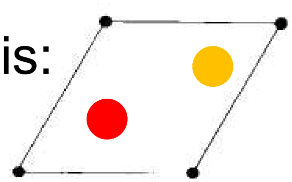


PRL 108, 155501 (2012)

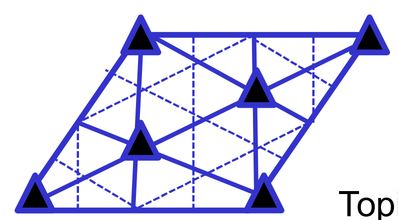
A and B atoms in the unit cell of silicene do not lie on the same plane. Consequently, the ground state structure of silicene lacks 6-fold rotational and two mirror plane symmetries which are present in the planar structure of graphene

Plane group: p3m1

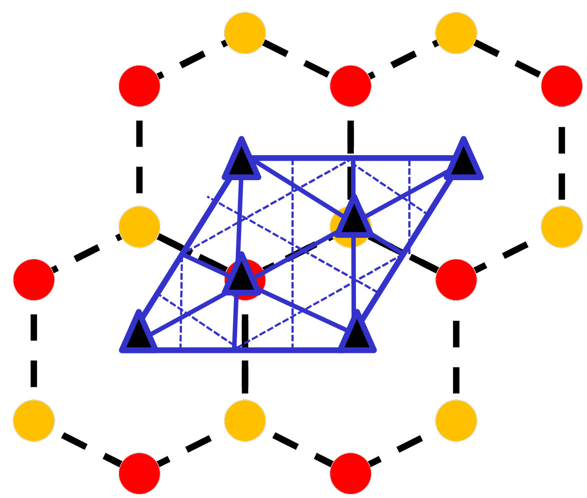
Basis:



The hexagonal *p*-lattice



2D Nanomaterial: Silicene



p3m1

SUMMARY

1D:

- The diad and reflection are the only symmetry operations allowed in 1D lattices
- There is one linear Bravais lattice
- Combination of reflection and lattice translation creates a glide line for the basis
- There are in total of 6 line groups in 1D

2D:

- The diad, triad, tetrad, hexad and reflection are the only symmetry operations allowed in 2D lattices
- There are 5 plane Bravais lattices
- Reflection and lattice translation create a glide line
- There are in total 17 plane groups in 2D

General:

- The line and space groups encode the Bravais lattice, the plane group and the principal symmetry elements consistent with the lattice <u>and</u> the basis